

Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices: exact results, modal scaling and inert defects

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Abstract

We study the first-passage time of a lazy nearest-neighbour random walk on the reflecting hypercubic lattice $\{0, \dots, N-1\}^d$ to a single absorbing site, mainly between opposite corners, and compare its mean T with its most probable value, the mode t^* . The means are exact finite sums; in two dimensions a single sum, equivalent to a known resistor-grid formula, is proved for every N , with the large- N expansion and closed-form coefficients. First-passage distributions are computed exactly by time stepping, where every mode is certified to be the global maximiser (in exact arithmetic, or with an a-priori bound on round-off), and by Laplace inversion to a relative accuracy of about 10^{-10} . In one dimension $t^*/T \rightarrow 0.33328$ for every activity q . For the three point-target placements studied the law is proved to be unimodal in continuous time in every dimension, and from corner to corner also in discrete time for $q \leq \frac{4}{5}$, a threshold attained on the chain with $N = 4$; unimodality fails near $q = 1$ and for other placements. For corner-to-corner transport in two and three dimensions a parameter-free closed form obtained from two-pole asymptotics, $t^* \approx T \ln(2dX)/(X + 2d - 1)$ with $X = \mu_1 q T$ and μ_1 the slowest relaxation rate of the lattice, has a relative error $O(1/(X \ln X))$ uniformly in $N \geq 12$ ($d = 2$) and $N \geq 10$ ($d = 3$), and the leading laws $t^* \sim N^2 \ln \ln N$ ($d = 2$), which the computed sizes do not reach, and $t^* \sim N^2 \ln N$ ($d = 3$), against $T \sim N^2 \ln N$ and $T \sim N^3$, hold with explicit remainders: the separation between mean and mode grows with dimension. These statements are proved in continuous time and, in discrete time, for $q \leq 0.99$ once N exceeds an explicit size and for every $q < 1$ once N exceeds an explicit q -dependent size; $q = 1$ is open. Several of these proofs are computer-assisted, with stated trusted bases and reproducible certificates. On the square lattice of side 35, uniformly random blocked sites up to a fraction 0.2 multiply mean and mode by nearly the same factor, the homogenised dilation of time D_0/D_{eff} , so that t^*/T changes by a few per cent at low density and by 4.7–8.9% at the fraction 0.2, whereas the placement of the blocked sites, not their number, decides the direction in which t^*/T changes.

Keywords: first-passage time; lattice random walk; reflecting boundaries; mean first-passage time; mode; unimodality; computer-assisted proof; effective resistance; inert defects.

1 Introduction

The time a confined random walker needs to reach a target, the first-passage time, is the basic quantity of search and reaction problems (Redner, 2001; Bénichou and Voituriez, 2014). Its distribution is usually summarised by its mean, but in bounded domains the mean need not be representative: the most probable value can lie far below it (Mattos et al., 2012; Godec and Metzler, 2016; Grebenkov et al., 2018). We study this gap in the simplest lattice setting. A lazy walker attempts, at each step and with probability q , a move to one of the $2d$ nearest neighbours on the hypercubic lattice $\Omega = \{0, \dots, N-1\}^d$; a move that would leave the lattice is cancelled, and one site is absorbing. We compare the mean first-passage time (MFPT) T with the mode t^* ,

the most probable first-passage time, mainly from one corner to the opposite corner, and ask how both depend on the size N , on the dimension $d = 1, 2, 3$ and, on the square lattice, on a fraction p of blocked sites (inert defects).

What is known. The generating-function formalism of [Montroll and Weiss \(1965\)](#) expresses the first-passage law through the propagator of the walk without target; with it [Montroll \(1969\)](#) found that the mean time to a single trap on a periodic lattice of n sites grows as $n \ln n$ in two dimensions and as n in three, and the survival probability of a walk on a finite lattice with a single trap was studied by [Weiss et al. \(1985\)](#). [Giuggioli \(2020\)](#) gave the exact propagator of the lazy walk in reflecting hypercubes of any dimension and, from it, the MFPT as a nested finite sum; for the rectangle, with the same boundary rule, the cosine double sum was given for this walk by [Condamin and Bénichou \(2006\)](#) (in resistance form already by [Wu \(2004\)](#)), and pseudo-Green-function methods give MFPTs in general confined domains ([Condamin et al., 2005, 2007a,b](#)). By the commute-time identity ([Chandra et al., 1989; Tetali, 1991](#)) mean hitting times of a symmetric walk are determined by effective resistances, and the resistance between two nodes of a rectangular grid of resistors with free boundaries is known as a double sum ([Wu, 2004](#)) and as a single sum for two arbitrary nodes and any grid size ([Tan and Tan, 2020](#)); the corner-to-corner resistance has a large-size expansion ([Essam and Wu, 2009](#)) whose coefficients are known for every aspect ratio ([Izmailian and Huang, 2010](#)).

The distribution in confinement is also known in outline. Rescaled by the mean it is close to exponential when exploration is non-compact ([Bénichou et al., 2010](#)), and for large volumes the rescaled laws fall into universality classes ([Meyer et al., 2011](#)); direct trajectories set a short time scale and trajectories that explore the whole domain a much longer one ([Godec and Metzler, 2016](#)); and for reversible Markov chains the exponential approximation of hitting times from a stationary start has explicit error bounds ([Aldous and Brown, 1992; Aldous and Fill, 2002](#)), and relaxation and first-passage time scales interlace ([Hartich and Godec, 2018, 2019](#)). For diffusion to a small target the slow decay rate follows from the perturbation of eigenvalues by a small removed region ([Ward and Keller, 1993](#)), the mean time to a small target on a reflecting boundary has an extensive asymptotic theory ([Holcman and Schuss, 2014](#)), and uniform approximations of the first-passage density combine that long-time behaviour with a short-time correction ([Isaacson and Newby, 2013](#)); the same authors observe that the most probable time, unlike the mean, depends strongly on the starting position, and compute it numerically. [Godec and Metzler \(2016\)](#) and [Grebenkov et al. \(2018\)](#) give the most probable time of direct trajectories, and [Grebenkov et al. \(2018\)](#) relate the median to the mean. [Sarvahaarman and Giuggioli \(2023\)](#) gave an analytic framework for lattice walks with inert heterogeneities and showed, for a single barrier on a chain, that its position decides whether the mean or only the mode and the tail respond; the reduction of conductivity and diffusivity on site-diluted lattices is a classical subject ([Kirkpatrick, 1973; Watson and Leath, 1974; Nieuwenhuizen et al., 1986; Ernst et al., 1987; Havlin and Ben-Avraham, 1987](#)).

Contributions. Each item states its status — proved (by hand or with computer assistance), computed exactly at the sizes stated (with no claim beyond them), or derived by formal asymptotics — and what was already known.

- (i) *Means (proved; known in substance).* A cosine sum for the MFPT between any two sites in any dimension (Theorem 4.1); in two dimensions single sums for opposite corners, in three equivalent forms, and for an arbitrary pair (Theorems 4.2 and 4.3), valid for every $N \geq 2$; the large- N expansion with closed-form coefficients (Theorem 4.5) and, with computer assistance, an explicit remainder for every N (Theorem 4.6); in three dimensions a double sum and, with computer assistance, $qT = 4.32046 \dots N^3 - 3.887 \dots N^2 + O(1)$, the leading constant being a sum of lattice Green's function values (Theorem 4.8). Through the commute-time identity the two-dimensional formulas are resistor-grid results in print

(Section 4.2); we add a self-contained proof in the random-walk setting and expressions of three expansion coefficients through the lemniscate constant, and record an overflow-free form of the single sum, an elementary rearrangement that is the one to use in floating point.

- (ii) *Distributions and modes (exact computation)*. First-passage distributions are computed by time stepping, by a closed form in one dimension and by Laplace inversion, and checked by a separately written pole expansion; a Cauchy–Schwarz bound, together with exact certificates for corner-to-corner transport at $q \in \{\frac{1}{2}, \frac{4}{5}, 1\}$ and an a-priori bound on round-off otherwise, certifies that every discrete-time mode obtained by time stepping on the lattice without defects is the global maximiser (Section 3); on lattices with defects the same certificate is evaluated in double precision (Section 8). In one dimension t^*/T tends to a constant; in two dimensions qt^*/N^2 increases at every computed size while t^*/T falls; in three dimensions t^* follows $N^2 \ln N$ whereas T grows as N^3 (Section 5). That the mode lies below the mean in confinement is known; the exact lattice values and their dependence on N and d are computed here.
- (iii) *A closed-form approximation of the mode (formal derivation)*. Two-pole asymptotics of the first-passage transform give a parameter-free formula for the mode in terms of $X = \mu_1 q T$, where μ_1 is the slowest relaxation rate of the reflecting lattice (Formal result 6.2); its accuracy is measured against the exact modes, and it fails in one dimension and for some placements of start and target. Its ingredients are the lattice counterparts of known continuum results (Ward and Keller, 1993; Isaacson and Newby, 2013). We have not found the explicit formula in the literature, have not established its absence, and claim no priority.
- (iv) *What is proved about the mode (theorems, several computer-assisted; Section 7.1)*. Section 7 proves, for corner-to-corner transport (end to end for $d = 1$) and, where stated, for the other point-target placements: that the law is unimodal for the three point-target placements studied in continuous time in every dimension and, in discrete time, for $q \leq \frac{4}{5}$ from corner to corner in every dimension (for the centre placements when $d \geq 2$), a threshold attained on the chain with $N = 4$, while unimodality fails near $q = 1$ on the chain, at $q = 1$ for some sizes in two dimensions, and for other placements; that the residues of the poles do not alternate in sign for $d \geq 2$, $N \geq 3$; that in one dimension $\lim t^*/T$ exists for every q and is enclosed to within 10^{-70} , with windows of width about one step for every $N \geq 3$ when $q \leq \frac{4}{5}$; and that in two and three dimensions the closed form has a scaled error $X\mu_1 |t_c^* - qt_{cf}|$ bounded for every $N \geq 12$ ($d = 2$) and $N \geq 10$ ($d = 3$), where t_{cf} is the closed form (35), and the leading laws $t^* \sim (4/\pi^2 q) N^2 \ln \ln N$ and $(6/\pi^2 q) N^2 \ln N$ hold with explicit remainders — in continuous time and, in discrete time, for $q \leq 0.99$ once N exceeds an explicit size and for every $q < 1$ once N exceeds an explicit size of order $\pi q/(1 - q)$ as $q \rightarrow 1$ (the case $q = 1$ is open) — and that the closed form is within 1% ($d = 2$) and 0.3% ($d = 3$) of the certified exact mode on finite ranges with $N \geq 10$ for $q \in \{\frac{4}{5}, \frac{1}{2}\}$, in three dimensions for every $N \geq 10$. Computer-assisted proofs reduce a statement, by lemmas proved by hand, to finitely many inequalities checked in exact or ball arithmetic; their trusted bases and certificates are named. Some ingredients are classical: the interlacing of the two spectra (Hartich and Godec, 2018, 2019), the mixture and exponential laws from the uniform and the quasi-stationary start (Aldous and Fill, 2002), and unimodality on the chain in continuous time (Keilson, 1971). We have not found the other statements in the literature, but our search was targeted, not exhaustive; the tools (interlacing, log-concavity under convolution, cone certificates) are classical.
- (v) *Shape of the distribution (exact computation)*. For $d \geq 2$ the law is close to an exponential preceded by a short dead time, whose end the mode marks; the mode is then not a typical time, and in three dimensions the maximum is very flat (Section 6.4). This two-scale

structure is known (Bénichou et al., 2010; Meyer et al., 2011; Godec and Metzler, 2016; Aldous and Brown, 1992); we add the exact lattice computation.

- (vi) *Inert defects (exact computation for each configuration; dilute law formal).* Each configuration of blocked sites on the square lattice is solved exactly, so the only uncertainty is the sampling of placements. At $N = 35$ and blocked fractions $p \leq 0.2$, uniformly random defects multiply the mean and the mode by nearly the same factor $D_0/D_{\text{eff}}(p)$ (within 2.8% and 1.82%, ensemble-mean estimates with 95% intervals up to $\pm 1.8\%$ and $\pm 0.72\%$), so that t^*/T changes by a few per cent at low density but rises by 4.7% to 8.9% at $p = 0.2$, and where the defects sit decides the direction of any change of t^*/T (Section 8); a single barrier on a chain already shows that placement decides which of the two times responds (Sarvaharman and Giuggioli, 2023). The dilute law $D_{\text{eff}}/D_0 = 1 - (\pi - 1)p + O(p^2)$ is in the literature (Watson and Leath, 1974; Nieuwenhuizen et al., 1986; Ernst et al., 1987); we prove the single-defect response behind it (Proposition 8.2). That the dilation becomes exact as $N \rightarrow \infty$ is stated only as Conjecture 8.4.

Outline. Section 2 defines the model and its exact structure; Section 3 describes the computations and the certificate of the global mode; Section 4 gives the exact means. Sections 5 and 6 report the exact modes and explain them through the poles of the first-passage transform, and Section 7 states what is proved about them. Section 8 treats inert defects, and Section 9 separates what is established from what is not. The Supplementary Material (Sections S1–S10) contains the proofs that are omitted here, the extended tables, the validation of the numerical methods and the description of the certificates; the companion notes R0–R5 in the repository (Zhouyi, 2026g) contain the complete proofs of the computer-assisted theorems.

2 Model, definitions and exact structure

This section fixes the model and collects the exact facts on which all later computations rest. The statements are elementary and largely standard; their proofs, a table of notation (Table S1) and numerical checks are in Supplementary Section S1.

2.1 The walk

Let $N \geq 2$, $d \geq 1$, $\Omega = \{0, \dots, N-1\}^d$ and $n = N^d$; $\mathbf{e}_1, \dots, \mathbf{e}_d$ are the lattice unit vectors. The definitions and the exact results hold in every dimension; the computations of this article are for $d \leq 3$, apart from a few finite ranges in dimensions 4 and 5 in Section 7. In each time step the walker attempts a move with probability $q \in (0, 1]$ (the *activity*) and rests otherwise. An attempted move goes to one of the $2d$ nearest neighbours, chosen uniformly, and is *cancelled* if it would leave Ω : the walker then stays where it is. With X_t the position after t steps, the transition matrix is

$$P(\mathbf{x}, \mathbf{x} \pm \mathbf{e}_i) = \frac{q}{2d} \quad \text{if } \mathbf{x} \pm \mathbf{e}_i \in \Omega, \quad P(\mathbf{x}, \mathbf{x}) = 1 - q + \frac{q}{2d} \eta(\mathbf{x}), \quad (1)$$

with all other entries zero, where $\eta(\mathbf{x})$ is the number of the $2d$ directions that lead out of Ω . P is symmetric, hence doubly stochastic with uniform stationary distribution, and irreducible; P_1 denotes the matrix with $q = 1$. The cancelled-move rule places mirrors half a lattice spacing outside the outermost sites; the bounce-back rule, which puts a walker that tries to leave on the interior neighbour, gives a non-symmetric matrix to which none of the formulas below applies.

2.2 First-passage quantities and the mode

Fix a start $\mathbf{o}$ and a target $\mathbf{a} \neq \mathbf{o}$, and let $\mathcal{T} = \min\{t \geq 0 : X_t = \mathbf{a}\}$. Let Q be P with the row and column of $\mathbf{a}$ removed and $\mathbf{r} = (I - Q)\mathbf{1}$ the vector of one-step absorption probabilities. By

irreducibility the spectral radius of Q is below one. With $\mathbf{e}_\mathbf{x}$ the coordinate vector of site $\mathbf{x}$, the survival probability is $S(t) = \mathbb{P}_\mathbf{o}(\mathcal{T} > t) = \mathbf{e}_\mathbf{o}^\top Q^t \mathbf{1}$, the first-passage probability mass function (PMF) is

$$f(t) = \mathbb{P}_\mathbf{o}(\mathcal{T} = t) = S(t-1) - S(t) = \mathbf{e}_\mathbf{o}^\top Q^{t-1} r, \quad t \geq 1, \quad (2)$$

and the mean first-passage time is

$$T = \mathbb{E}_\mathbf{o}[\mathcal{T}] = \sum_{t \geq 0} S(t) = \mathbf{e}_\mathbf{o}^\top (I - Q)^{-1} \mathbf{1}. \quad (3)$$

We write $T_{\mathbf{o} \rightarrow \mathbf{a}}$ when the pair matters. For a target set (used only for the exit geometry of Section 2.6) the rows and columns of all target sites are removed, and everything below holds unchanged.

Definition 2.1 (Mode). The mode, or most probable first-passage time, is the smallest maximiser of the exact PMF,

$$t^* = \min \left\{ t \geq 1 : f(t) = \max_{\tau \geq 1} f(\tau) \right\},$$

and $\varrho = t^*/T$ is the mode-to-mean ratio.

The definition involves no smoothing, fit or acceptance window, and it does not assume that f has a single local maximum; when that holds is the subject of Section 7.3.

Continuous time. The companion walk performs the jumps of P_1 at the events of a Poisson process of rate q ; its generator is $P - I$ and its first-passage density is $g_q(t) = \mathbf{e}_\mathbf{o}^\top e^{-(I-Q)t} r$. We write $g = g_1$ for the unit-rate density and t_c^* for its smallest maximiser on $[0, \infty)$.

2.3 Activity, the two time conventions and the spectral form

Let Q_1 and r_1 be Q and r at $q = 1$. For each distinct eigenvalue ν of the symmetric matrix $I - Q_1$, which lies in $(0, 2)$, let Π_ν be the orthogonal projection onto the eigenspace and $b_\nu = \mathbf{e}_\mathbf{o}^\top \Pi_\nu r_1$; list the eigenvalues with $b_\nu \neq 0$ as $\nu_0 < \nu_1 < \dots < \nu_J$ and write $b_j = b_{\nu_j}$. Let $\tilde{F}_q(z) = \sum_{t \geq 1} f(t) z^t$ and $\hat{F}(s) = \int_0^\infty e^{-st} g(t) dt$.

Proposition 2.2 (Exact structure). *For every $N \geq 2$, $d \geq 1$, $q \in (0, 1]$ and $\mathbf{o} \neq \mathbf{a}$:*

(a) $I - P = q(I - P_1)$, $I - Q = q(I - Q_1)$ and $r = q r_1$. Hence

$$qT \text{ does not depend on } q. \quad (4)$$

(b) For every $z \neq 0$ at which $I - zQ$ is invertible, in particular for $0 < |z| \leq 1$,

$$\tilde{F}_q(z) = \hat{F}\left(\frac{1-z}{qz}\right). \quad (5)$$

(c) The PMF and the unit-rate density are finite sums with the same rates and residues,

$$g(t) = \sum_{j=0}^J b_j e^{-\nu_j t}, \quad f(t) = \sum_{j=0}^J q b_j (1 - q\nu_j)^{t-1}, \quad \hat{F}(s) = \sum_{j=0}^J \frac{b_j}{s + \nu_j}, \quad (6)$$

with $\sum_j b_j / \nu_j = 1$, $\sum_j b_j / \nu_j^2 = qT$ and $S(t) = \sum_j (b_j / \nu_j) (1 - q\nu_j)^t$.

(d) $g_q(t) = q g(qt)$: the continuous-time walk at activity q has mode exactly t_c^*/q and mean exactly T . If $q \leq \frac{1}{2}$, all $1 - q\nu_j > 0$ and no term of f alternates in sign; in general $1 - q\nu_j > 1 - 2q$.

The discrete-time and continuous-time problems are therefore one problem: inverting the generating function and inverting the Laplace transform recover the same rates ν_j and residues b_j . The products qT and q times the continuous-time mode at activity q do not depend on q ; the discrete mode t^* is not exactly t_c^*/q , but at $q = 0.8$ the two differ by at most 3.05 steps in the 54 cases compared (Supplementary Section S2.2).

2.4 Renewal ratio and the two spectra

Two spectra enter the problem. The first belongs to the reflecting lattice *without* a target: the unit-rate propagator has the Laplace transform (Giuggioli, 2020)

$$\hat{P}(\mathbf{x}, s \mid \mathbf{y}) = [(sI + I - P_1)^{-1}]_{\mathbf{xy}} = \sum_{\mathbf{k}} \frac{\phi_{\mathbf{k}}(\mathbf{x}) \phi_{\mathbf{k}}(\mathbf{y})}{s + \mu_{\mathbf{k}}}, \quad \phi_{\mathbf{k}}(\mathbf{x}) = \prod_{i=1}^d \psi_{k_i}(x_i), \quad (7)$$

with $\psi_k(m) = \sqrt{\alpha_k/N} \cos \vartheta_k(m)$, $\mathbf{k} \in \{0, \dots, N-1\}^d$, and the *reflecting rates* $\mu_{\mathbf{k}} = \frac{2}{d} \sum_i s_{k_i}^2$ (Lemma S3.1); here $\vartheta_k(m) = \pi k(2m+1)/(2N)$, $s_k = \sin(\pi k/2N)$, $\alpha_0 = 1$ and $\alpha_k = 2$ for $k \geq 1$. The smallest non-zero rate is

$$\mu_1 = \frac{2}{d} \sin^2 \frac{\pi}{2N} = \frac{\pi^2}{2dN^2} [1 + O(N^{-2})]. \quad (8)$$

The second spectrum, $\{\nu_j\}$, belongs to the walk *with* the target; for $d \geq 2$ it is not known in closed form. The two are linked by the renewal ratio (Montroll and Weiss, 1965; Redner, 2001)

$$\hat{F}(s) = \frac{\hat{P}(\mathbf{a}, s \mid \mathbf{o})}{\hat{P}(\mathbf{a}, s \mid \mathbf{a})}, \quad (9)$$

and the same ratio of discrete-time propagators equals $\tilde{F}_q(z)$.

Proposition 2.3 (Interlacing of the two spectra). *Fix the target $\mathbf{a}$. Let $0 = \mu_{(0)} < \mu_{(1)} < \dots < \mu_{(m)}$ be the distinct reflecting rates whose eigenspaces have non-zero weight $w_i = \sum_{\mathbf{k}: \mu_{\mathbf{k}} = \mu_{(i)}} \phi_{\mathbf{k}}(\mathbf{a})^2$ at the target.*

(i) $\hat{P}(\mathbf{a}, s \mid \mathbf{a})$ has exactly m zeros in the complex plane. They are real and simple, $s = -\nu_{(i)}$, and

$$0 < \nu_{(0)} < \mu_{(1)} < \nu_{(1)} < \mu_{(2)} < \dots < \nu_{(m-1)} < \mu_{(m)}. \quad (10)$$

(ii) The $\nu_{(i)}$ are exactly the eigenvalues ν of $I - Q_1$ with $\Pi_{\nu} r_1 \neq 0$, and every pole of $\hat{F}$ is a zero of $\hat{P}(\mathbf{a}, s \mid \mathbf{a})$.

(iii) If the sites other than $\mathbf{a}$ form a connected set (always for $d \geq 2$), then for every start $\nu_0 = \nu_{(0)}$ is the smallest eigenvalue of $I - Q_1$, it is simple, $b_0 > 0$, and

$$\frac{1}{\nu_0} = \frac{\sum_{\mathbf{x} \neq \mathbf{a}} v(\mathbf{x}) q T_{\mathbf{x} \rightarrow \mathbf{a}}}{\sum_{\mathbf{x} \neq \mathbf{a}} v(\mathbf{x})}, \quad (11)$$

where $v > 0$ is the corresponding eigenvector (the quasi-stationary distribution).

The interlacing of relaxation and first-passage time scales holds for reversible Markov dynamics in general (Hartich and Godec, 2018, 2019), and (11) expresses the classical fact that from the quasi-stationary distribution the first-passage time is exponential with rate ν_0 (Aldous and Fill, 2002); Proposition 2.3 collects these facts for the present walk.

Two consequences are used throughout. First, the long-time decay of the survival probability is set by $\nu_0 < \mu_{(1)}$ (for a corner target $\mu_{(1)} = \mu_1$), not by the reflecting gap, and (11) makes $1/\nu_0$ an average of mean first-passage times: for corner-to-corner transport $\nu_0 qT$ falls from 1.164 ($N = 5$) to 1.021 ($N = 16\,777\,216$) in two dimensions and from 1.076 ($N = 5$) to 1.0001 ($N = 4096$) in three, in agreement with the exponential approximation of hitting times in reversible chains (Aldous and Brown, 1992; Aldous and Fill, 2002). Second, the separation of the two spectra is measured by

$$X = \mu_1 qT,$$

the mean first-passage time in units of the relaxation time of the box: $X = 22.706$ for $d = 2$, $N = 35$ and $X = 277.739$ for $d = 3$, $N = 40$. For $d \geq 2$ the target is a weak sink on the diffusive time scale ($X \gg 1$), and the closed form of Section 6 expresses t^*/T through X and d alone.

2.5 Mean first-passage time and effective resistance

Proposition 2.4 (Mean hitting time and pseudo-inverse). *Let P be a symmetric, irreducible stochastic matrix on n states, $L = I - P$ and L^+ its Moore–Penrose pseudo-inverse. For a target state $\mathbf{a}$, $\mathbb{E}_{\mathbf{x}}[\mathcal{T}]$ is the unique function h with $h(\mathbf{a}) = 0$ and $(Lh)(\mathbf{x}) = 1$ for $\mathbf{x} \neq \mathbf{a}$, and*

$$T_{\mathbf{x} \rightarrow \mathbf{a}} = n (L_{\mathbf{aa}}^+ - L_{\mathbf{xa}}^+). \quad (12)$$

This is the standard fundamental-matrix, or pseudo-Green-function, expression for hitting times (Aldous and Fill, 2002; Condamin et al., 2005, 2007a). Since $I - P = \frac{q}{2d} \mathcal{L}$, where $\mathcal{L}$ is the Laplacian of the grid graph with unit conductance on every nearest-neighbour bond, $G = \mathcal{L}^+$ gives

$$T_{\mathbf{x} \rightarrow \mathbf{a}} = \frac{2dn}{q} (G_{\mathbf{aa}} - G_{\mathbf{xa}}), \quad T_{\mathbf{x} \rightarrow \mathbf{a}} + T_{\mathbf{a} \rightarrow \mathbf{x}} = \frac{2dn}{q} R(\mathbf{x}, \mathbf{a}), \quad (13)$$

where R is the effective resistance between the two nodes (Klein and Randić, 1993); the second relation is the commute-time identity (Chandra et al., 1989; Tetali, 1991; Levin et al., 2009). The same argument applies on any connected set of sites and is used for lattices with blocked sites in Section 8. Equivalently, $T = \tilde{F}'(1) = \lim_{z \rightarrow 1^-} (1 - \tilde{F}(z))/(1 - z)$; the pole of the propagator at $z = 1$ (the zero mode) cancels in the renewal ratio (Supplementary Section S1).

2.6 Geometries

Four placements of start and target are used. CC (corner to corner): $\mathbf{o} = (0, \dots, 0)$ and $\mathbf{a} = (N - 1, \dots, N - 1)$; this is the default, and for $d = 1$ it is the passage from the reflecting end of a chain to its other end. C2M: corner start, target at the centre $(\frac{N-1}{2}, \dots, \frac{N-1}{2})$. M2C: centre start, corner target. EXIT: centre start, with every site of the boundary layer absorbing. The last three require odd N . Unless stated otherwise, discrete-time values are for $q = 0.8$ and continuous-time values for unit rate; times are in steps.

3 Exact computation of first-passage distributions

Every mode reported in this article is the maximiser of a computed first-passage distribution; no smoothing, acceptance window or fitted parameter enters. For time stepping and for the closed form of the chain (routes A and B below) “exact” means exact up to double-precision round-off; for Laplace inversion (route C) it means numerically exact to a measured accuracy: the position of the mode is reproduced by a separately written route to 1.1×10^{-10} or better. The implementation, the checks of Table S2, a Monte Carlo simulation of the walker rule and a demonstration of the pitfalls of inverting a generating function numerically are in Supplementary Section S2.

3.1 Four routes

Route A (time stepping). The vector $\rho_t(\mathbf{x}) = \mathbb{P}_{\mathbf{o}}(X_t = \mathbf{x}, \mathcal{T} > t)$ obeys $\rho_{t+1} = Q\rho_t$, a nearest-neighbour stencil of cost $O(N^d)$ per step, and $f(t+1) = \sum_{\mathbf{x}} \rho_t(\mathbf{x})P(\mathbf{x}, \mathbf{a})$. The data set consists of 193 runs: the chain up to $N = 1000$, CC in two dimensions up to $N = 401$ and in three dimensions up to $N = 100$ (10^6 sites), the geometries C2M, M2C and EXIT, and activities $q \in \{0.3, 0.5, 0.8, 0.9, 1\}$. *Route B (one dimension).* For the chain the PMF has the closed form (29); for large N the integer maximiser is located from the sign of $f(t+1) - f(t)$, with the logarithm of each eigenvalue formed as $\log_{1p}(-q\nu_m)$. *Route C (Laplace space).* We evaluate the renewal ratio (9) at unit rate, with one of the d spectral sums of $\hat{P}$ carried out in closed form (Lemma S2.1), and invert it with the fixed-Talbot rule (Talbot, 1979; Abate and Valkó, 2004; Weideman, 2006); the mode t_c^* is the root of g' , bracketed to a relative width below 10^{-11} . Against the exact density, written as the Erlang mixture of the PMF at $q = 1$, the median relative

error of the inversion is 1.4×10^{-13} and the largest 5.4×10^{-11} . This route reaches $N = 16\,777\,216$ for $d = 2$ and $N = 4096$ for $d = 3$. *Route P (pole expansion)*. A second implementation, written separately and sharing with route C only the spectral form of the propagator, finds the poles $-\nu_j$ of $\hat{F}$ as the zeros of $\hat{P}(\mathbf{a}, s \mid \mathbf{a})$, one between consecutive reflecting rates (Proposition 2.3), and maximises the finite sum (6) directly. For corner-to-corner transport the continuous-time modes of routes C and P agree to 7×10^{-13} at all 40 sizes compared for $d = 2$ and to 1.1×10^{-10} at all 24 sizes for $d = 3$; the weaker agreement in three dimensions reflects how flat the density is near its maximum. A second time-stepping code reproduces every integer mode in the 148 cases it was run on.

Routes A and B work in discrete time at activity q , routes C and P in continuous time at unit rate. In the 54 cases computed by both at $q = 0.8$, $t^* = t_c^*/q + c$ with $-3.05 \leq c \leq -0.22$ steps; the relative difference is below 10^{-4} for all $N \geq 100$. At $q = 1$ parity is visible (the PMF of the chain lies on two branches); the discrete-time results below are for $q = 0.8$, where it is not, unless another activity is stated.

3.2 Certificate of the global mode

Definition 2.1 asks for the maximiser of f over all $t \geq 1$, whereas a computation covers a finite range. The gap is closed by a bound that uses only the symmetry of Q ; it therefore holds on the clean lattice, for extended targets, and on the lattices with blocked sites of Section 8.

Proposition 3.1 (Certificate of the global mode). *Let P be a symmetric stochastic matrix, Q the matrix obtained by deleting the rows and columns of the target states, $r = (I - Q)\mathbf{1}$ and $f(t) = \mathbf{e}_o^\top Q^{t-1}r$. Let λ_k and v_k be the eigenvalues and orthonormal eigenvectors of Q and $w_k = v_k(\mathbf{o}) (v_k^\top r)$, so that $f(t) = \sum_k w_k \lambda_k^{t-1}$, and put $\mathcal{E}(t) = \sum_k |w_k| |\lambda_k|^{t-1}$. Then:*

- (a) $\mathcal{E}$ is non-increasing and $f(t') \leq \mathcal{E}(t)$ for all $t' \geq t \geq 1$;
- (b) for all integers $j, l \geq 0$ and all $t' \geq j + l + 1$, $f(t') \leq \|Q^j \mathbf{e}_o\| \|Q^l r\|$ (Euclidean norms);
- (c) if $t_{\text{cert}} \geq 2$ and $\mathcal{E}(t_{\text{cert}})$, or the bound of (b) for some $j + l + 1 = t_{\text{cert}}$, is smaller than $\max_{1 \leq t < t_{\text{cert}}} f(t)$, then the mode of Definition 2.1 is the smallest maximiser of f on $\{1, \dots, t_{\text{cert}} - 1\}$.

Proof. Q is symmetric with non-negative entries and row sums at most one, so $\|Q\|_2 \leq 1$ and $|\lambda_k| \leq 1$. (a) For $t' \geq t$, $|\lambda_k|^{t'-1} \leq |\lambda_k|^{t-1}$. (b) Write $t' - 1 = j' + l'$ with $j' \geq j$ and $l' \geq l$; by (2), the symmetry of Q and the Cauchy–Schwarz inequality, $f(t') = (Q^{j'} \mathbf{e}_o)^\top (Q^{l'} r) \leq \|Q^{j'} \mathbf{e}_o\| \|Q^{l'} r\|$. (c) By (a) or (b), $f(t') < \max_{1 \leq t < t_{\text{cert}}} f(t)$ for every $t' \geq t_{\text{cert}}$. $\square$

We applied bound (b) with $j = l = k$ to every run of route A, with a third time-stepping code that advances $u_k = Q^k \mathbf{e}_o$ and $v_k = Q^k r$ alternately and stops at the first k with $\|u_k\| \|v_k\| < (1 - 10^{-6}) \max_{t \leq 2k} f(t)$. All 193 runs are certified, and every mode equals the one found by route A; for corner-to-corner transport at $q = 0.8$ and $N \geq 20$ the certificate is reached at t_{cert}/t^* between 1.40 and 1.53 for the chain, 1.25 and 1.32 for $d = 2$, and 1.15 and 1.19 for $d = 3$. Below t_{cert} the maximiser is found by comparing computed values. For the 97 corner-to-corner runs at $q \in \{0.5, 0.8, 1\}$ the sizes lie in the ranges in which the companion note R5 certifies the mode in exact arithmetic (Theorems 7.9 and 7.15; for the chain at $q = 1$, [R5, Thms 13, 40]), and all 97 modes agree with the certified ones. For the other 96 runs we use an a-priori bound of the round-off of the stepping: all its operations act on non-negative numbers, so every computed value of $f(t)$ lies within the factors $(1 \pm \delta)^t$ of the exact value (with a further factor $(1 \pm u)^{m-1}$, $u = 2^{-53}$, for m target sites), where $\delta \leq 1.0 \times 10^{-15}$ per step, and in each of these runs every other computed value, inflated by this factor, stays below the deflated maximum (the smallest relative gap between the maximum and any other value is 6.59×10^{-11} ; Supplementary Section S2.3). Every discrete-time mode of Sections 3–7 (the 193 runs of route A on the lattice

without defects) — in particular all 81 corner-to-corner modes behind Table 2 — is therefore the global maximiser of its PMF. The modes of the lattices with blocked sites of Section 8 are certified by the same proposition, but in double precision and without this a-priori bound on round-off. That the PMF has a single local maximum is not assumed; for the placements studied it is proved in Section 7.3, so that the single sign change of g' found by route C at sizes beyond time stepping is the global maximiser as well.

4 Exact mean first-passage times

By (4) the product qT does not depend on q , so all results are stated for qT . The proofs are in Supplementary Sections S3 and S4 (Section S8 for the two computer-assisted additions), together with the numerical verification; only their idea is given here. The two-dimensional formulas are known in substance; Section 4.2 says which formula is in which paper.

4.1 A cosine sum in any dimension and single sums in two dimensions

Theorem 4.1 (Cosine sum). *For the walk of Section 2 on $\Omega = \{0, \dots, N-1\}^d$, any start $\mathbf{o}$ and target $\mathbf{a}$, and every $q \in (0, 1]$,*

$$qT_{\mathbf{o} \rightarrow \mathbf{a}} = \frac{d}{2} \sum_{\mathbf{k} \neq \mathbf{0}} \left(\prod_{i=1}^d \alpha_{k_i} \right) \frac{\prod_i \cos^2 \vartheta_{k_i}(a_i) - \prod_i \cos \vartheta_{k_i}(o_i) \cos \vartheta_{k_i}(a_i)}{\sum_{i=1}^d s_{k_i}^2}, \quad (14)$$

where $\mathbf{k}$ runs over $\{0, \dots, N-1\}^d$, $\vartheta_k(m) = \pi k(2m+1)/(2N)$, $s_k = \sin(\pi k/2N)$, $\alpha_0 = 1$ and $\alpha_k = 2$ for $k \geq 1$. For the corner start $\mathbf{o} = (0, \dots, 0)$ and the opposite corner $\mathbf{a} = (N-1, \dots, N-1)$,

$$qT_N^{(d)} = d \sum_{\substack{\mathbf{k} \in \{0, \dots, N-1\}^d \\ k_1 + \dots + k_d \text{ odd}}} \frac{\prod_i \alpha_{k_i} c_{k_i}^2}{\sum_i s_{k_i}^2}, \quad c_k = \cos \frac{\pi k}{2N}. \quad (15)$$

In one dimension $T_N^{(1)} = N(N-1)/q$, and more generally $qT_{m \rightarrow m'} = h_1(m, m')$ with

$$h_1(m, m') = \begin{cases} (m' - m)(m' + m + 1), & m \leq m', \\ (m - m')(2N - 1 - m - m'), & m \geq m'. \end{cases} \quad (16)$$

The proof inserts the spectral form of the pseudo-inverse of Proposition 2.4; for opposite corners $\cos \vartheta_k(N-1) = (-1)^k c_k$, so only the modes with $k_1 + \dots + k_d$ odd survive. For $d = 2$ one of the two indices can be summed in closed form.

Theorem 4.2 (Single sums for opposite corners, $d = 2$). *Let $\sigma_k = 2 - \cos(\pi k/N) = 1 + 2s_k^2$ and $\varphi_k = \operatorname{arcosh} \sigma_k = 2 \operatorname{arsinh} s_k$. For every $N \geq 2$ and $q \in (0, 1]$:*

(a)

$$qT_N = 2N(N-1) + 4N \sum_{k=1}^{N-1} \frac{c_k^2 \mathcal{B}_k}{\sinh \varphi_k \sinh N\varphi_k}, \quad (17)$$

$$\mathcal{B}_k = \cosh N\varphi_k + \cosh(N-1)\varphi_k - (-1)^k [1 + \cosh \varphi_k];$$

(b) (overflow-free form, all terms positive)

$$qT_N = 4N \sum_{k=1}^{N-1} \cos^2 \frac{\pi k}{2N} \frac{\sqrt{1 + s_k^2}}{s_k} g_k, \quad g_k = \begin{cases} \coth(N \operatorname{arsinh} s_k), & k \text{ odd}, \\ \tanh(N \operatorname{arsinh} s_k), & k \text{ even}, \end{cases} \quad (18)$$

(c) (half-range sums) with

$$S_{\text{odd}} = \sum_{\substack{1 \leq k \leq N-1 \\ k \text{ odd}}} c_k^2 \coth \frac{\varphi_k}{2} \coth \frac{N\varphi_k}{2}, \quad S_{\text{even}} = \sum_{\substack{2 \leq k \leq N-1 \\ k \text{ even}}} c_k^2 \coth \frac{\varphi_k}{2} \tanh \frac{N\varphi_k}{2}, \quad (19)$$

one has

$$q T_N = 8N S_{\text{odd}} - 2N^2 = 2N^2 + 8N S_{\text{even}}, \quad S_{\text{odd}} - S_{\text{even}} = \frac{N}{2}. \quad (20)$$

Idea of the proof. In (15) the row $k = 0$ is a one-dimensional hitting time and contributes $2N(N-1)$. For $k \geq 1$ the sum over the second index is the Green's function of a ring of $2N$ sites, which is elementary (Lemma S3.2); this gives (a), and half-angle identities turn (a) into (b) and (c). Exact rational solves of $(I - Q)h = \mathbf{1}$ for $N = 2, \dots, 40$ (84 cases) agree with all forms, evaluated with 60 digits, to a relative deviation of at most 6.6×10^{-60} (Supplementary Section S5.1). In double precision form (a) returns NaN for every $N \geq 403$, because $\cosh N\varphi_k$ overflows, whereas form (b) contains only $\tanh$ and $\coth$ and agrees with 40-digit values to 1.4×10^{-16} or better at the five sizes tested between $N = 10$ and $N = 6000$.

Theorem 4.3 (Single sum for an arbitrary start and target, $d = 2$). *For all $\mathbf{o} = (o_1, o_2)$ and $\mathbf{a} = (a_1, a_2)$ in $\{0, \dots, N-1\}^2$,*

$$q T_{\mathbf{o} \rightarrow \mathbf{a}} = 2 h_1(o_2, a_2) + 4N \sum_{k=1}^{N-1} \left[\mathcal{C}_k(a_1, a_1) \Psi_k(a_2, a_2) - \mathcal{C}_k(o_1, a_1) \Psi_k(o_2, a_2) \right], \quad (21)$$

with $\mathcal{C}_k(m, m') = \cos \vartheta_k(m) \cos \vartheta_k(m')$, h_1 as in (16), and

$$\Psi_k(m, m') = \frac{\cosh((N - |m - m'|)\varphi_k) + \cosh((N - 1 - m - m')\varphi_k)}{\sinh \varphi_k \sinh N\varphi_k}. \quad (22)$$

Against exact rational solves, (21) holds to 8.0×10^{-40} (40-digit arithmetic) for all 924 ordered pairs with $N \leq 5$ at two values of q and for 72 random pairs with $6 \leq N \leq 30$.

Corollary 4.4 (Effective resistance). *Let $R(\mathbf{x}, \mathbf{y})$ be the effective resistance between two nodes of the $N \times N$ grid graph with unit resistors on its edges, and R_N its value for opposite corners. Then*

$$T_{\mathbf{o} \rightarrow \mathbf{a}} + T_{\mathbf{a} \rightarrow \mathbf{o}} = \frac{4N^2}{q} R(\mathbf{o}, \mathbf{a}), \quad q T_N = 2N^2 R_N, \quad (23)$$

so that $R_N = 1 + \frac{4}{N} S_{\text{even}} = \frac{4}{N} S_{\text{odd}} - 1$.

4.2 Relation to the literature

Through (23) the mean first-passage time between two sites is a resistance of the $N \times N$ resistor grid with free boundaries, which is a solved problem. The cosine double sum, (14) with $d = 2$, was given by [Condamine and Bénichou \(2006\)](#) for the same walk with the same boundary rule; in resistance form it is Eq. (37) of [Wu \(2004\)](#), and it is the two-dimensional case of the nested sums of [Giuggioli \(2020\)](#). The half-sum S_{even} is the formula of [Essam and Wu \(2009\)](#) for the corner-to-corner resistance, stated there for even sizes. Form (a) of Theorem 4.2, for both parities of N , is Eq. (64) of [Tan and Tan \(2020\)](#) for the regular rectangle, evaluated for unit resistors, and their Eq. (57), a single sum for the resistance between two arbitrary nodes, coincides with the commute time obtained from Theorem 4.3: evaluated as printed, Eq. (64) agrees with form (a) to 2.6×10^{-38} in every term for twelve sizes between $N = 2$ and $N = 101$, and Eq. (57) agrees with Theorem 4.3 to 3.2×10^{-40} for 964 pairs. Single-sum resistance formulas for other networks are related work, not sources of the present formulas ([Izmailian et al., 2014](#); [Tan, 2015](#);

Essam et al., 2015). The single sums are therefore in print, in resistance form; form (b) and the odd half-sum of (c) are elementary rearrangements of them, which we have not seen printed in this form and for which we claim no novelty. What is added is a self-contained elementary proof in the random-walk setting for both parities of N ; we also record form (b), an elementary rearrangement, because it is the one to use in floating-point arithmetic. The relation (23) between the lazy reflecting walk and the resistor grid is the classical commute-time identity (Chandra et al., 1989; Tetali, 1991).

4.3 Large- N expansion in two dimensions

Theorem 4.5 (Large- N expansion, $d = 2$). *Let $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.6220575542\dots$ be the lemniscate constant, γ_E Euler’s constant and $\Xi = \varpi^4/\pi^2$ ($= 4.7892679494\dots$). For every fixed integer $M \geq 1$, as $N \rightarrow \infty$,*

$$qT_N = \frac{8}{\pi}N^2 \ln N + C_2 N^2 + \sum_{m=1}^M C_{2-2m} N^{2-2m} + O(N^{-2M}), \quad (24)$$

with constants C_{2-2m} given for every $m \geq 0$ by the finite expressions (S4.14) of Proposition S4.3; in particular

$$\begin{aligned} C_2 &= \frac{8}{\pi} \left[\gamma_E + \ln \frac{4\sqrt{2}}{\varpi} - \frac{1}{2} - \frac{\pi}{4} \right] = 0.154637787818917\dots, \\ C_0 &= \frac{\Xi}{9} = 0.532140883276956\dots, \quad C_{-2} = -\frac{\pi \Xi (25\Xi + 648)}{10800} = -1.069558947686132\dots \end{aligned} \quad (25)$$

Only even powers of N occur; there is no $N \ln N$ term and no term linear in N .

Idea of the proof (Supplementary Section S4). In form (b) write $g_k = 1 + (g_k - 1)$. With $g_k = 1$ the sum is a Riemann sum of a function with a simple pole at the origin, and the Euler–Maclaurin formula gives $(8/\pi)N^2 \ln N$, a constant times N^2 and even powers of N . The corrections $g_k - 1$ are exponentially small unless $k = O(\ln N)$; expanding there produces Lambert series at the point $-e^{-\pi}$, which are values of Eisenstein series at the lemniscatic point. This is where ϖ comes from.

Two terms of (24) are accurate to 4.7×10^{-5} at $N = 35$ and to 3.0×10^{-8} at $N = 1000$, and five terms reach 1.2×10^{-16} at $N = 100$. Closed forms of C_{-4} , C_{-6} and C_{-8} , obtained by computer algebra, agree with a Richardson extrapolation of exact values to within 3×10^{-33} , 8×10^{-25} and 2×10^{-16} , and a blind linear solve for the coefficients of twelve even powers from 70-digit exact values recovers $C_2, \dots, C_{-8}$ to between 67 and 25 digits (Table 1). Essam and Wu (2009) obtained $R_N = \frac{4}{\pi} \ln N + 0.077318\dots + 0.266070\dots N^{-2} - 0.534779\dots N^{-4} + \dots$, with the last two constants given numerically; $C_2/2$, $C_0/2$ and $C_{-2}/2$ agree with every digit they print, and Eqs. (42)–(43) of Izmailian and Huang (2010), evaluated at aspect ratio one, reproduce $C_2/2$ and $C_0/2$ to 50 digits. The closed forms (25) are therefore a specialisation of known results; to our knowledge they have not been printed in terms of ϖ .

Theorem 4.5 controls the remainder only asymptotically. A bound valid for every N is computer-assisted (the status words are defined in Section 7.1).

Theorem 4.6 (Explicit remainder for every N ; computer-assisted). *Let $r_N = qT_N - \frac{8}{\pi}N^2 \ln N - C_2 N^2$. For every $N \geq 2$, $-11.2 \leq r_N \leq 11.25$. For $2 \leq N \leq 1500$, r_N is strictly increasing; $0.3211 < r_N < 0.53213$ for $N \leq 301$ and $r_N < 0.5321405 < C_0$ for $301 < N \leq 1500$. (Source: [R2, Thm 9.1], [R5, Thm 26, Prop 27], bases T3 and T1.)*

The target on the diffusive scale. Section 6 uses $X = \mu_1 qT$, here with $\mu_1 = \sin^2(\pi/2N)$. By Theorem 4.5,

$$X = \mu_1 qT_N = 2\pi \ln N + \frac{\pi^2}{4} C_2 + O(N^{-2} \ln N) = 2\pi \ln N + 0.381553\dots + O(N^{-2} \ln N), \quad (26)$$

and for every $N \geq 2$ the error is at most $(33.0 + 5.2 \ln N)/N^2$ in modulus (computer-assisted; [R2, Cors 11.1, 11.2]).

4.4 Three dimensions

Proposition 4.7 (Double sum for opposite corners, $d = 3$). *Let $\varphi_{kj} = 2 \operatorname{arsinh} \sqrt{s_k^2 + s_j^2}$. For every $N \geq 2$ and $q \in (0, 1]$,*

$$q T_N^{(3)} = 3N \left[\sum_{\substack{0 \leq k, j \leq N-1 \\ (k, j) \neq (0, 0)}} \alpha_k \alpha_j c_k^2 c_j^2 \coth \frac{\varphi_{kj}}{2} g_{kj} - N(N-1) \right], \quad (27)$$

where $g_{kj} = \coth(N\varphi_{kj}/2)$ if $k + j$ is odd and $g_{kj} = \tanh(N\varphi_{kj}/2)$ if $k + j$ is even.

It gives, for example, $q T_N^{(3)}/N^3 = 4.281589, 4.316573$ and 4.319511 at $N = 100, 1000$ and 4096 .

Theorem 4.8 (Leading constants, $d = 3$; computer-assisted). *Let $\mathcal{G}_\infty(\mathbf{x})$ be the Green's function of the unit-rate walk on $\mathbb{Z}^3$, the expected time spent at $\mathbf{x}$ by a walk started at the origin ($\mathcal{G}_\infty(0, 0, 0) = 1.5163860592$ is Watson's integral; [Watson, 1939](#)). For every $N \geq 2$,*

$$q T_N^{(3)} = K_3 N^3 + K'_3 N^2 + R(N), \quad -99.8 \leq R(N) \leq 99.8,$$

where

$$\begin{aligned} K_3 &= \mathcal{G}_\infty(0, 0, 0) + 3\mathcal{G}_\infty(1, 0, 0) + 3\mathcal{G}_\infty(1, 1, 0) + \mathcal{G}_\infty(1, 1, 1), \\ 4.320460169373 &< K_3 < 4.320460169375, \\ K'_3 &= c_{\tau 3} + c_{m\tau 3} = -3.887254142500 \dots, \end{aligned}$$

with the lattice series

$$\begin{aligned} c_{\tau 3} &= 2 + \frac{12}{\pi} (\gamma_E - \ln \pi - 2 \ln 2) + 3E_\infty, & c_{m\tau 3} &= 1 - \frac{6}{\pi} \sum_{\mathbf{k} \in \mathbb{Z}^2 \setminus 0} \frac{(-1)^{k_1+k_2}}{|\mathbf{k}| \sinh(\pi|\mathbf{k}|)}, \\ E_\infty &= \frac{4}{\pi} \sum_{\mathbf{k} \in \mathbb{Z}^2 \setminus 0} F_*(|\mathbf{k}|) + \frac{16}{\pi} \sum_{j \geq 1} \int_0^\infty F_*(\sqrt{j^2 + v^2}) dv, & F_*(x) &= \frac{1}{x(e^{2\pi x} - 1)}. \end{aligned}$$

Moreover $R(N) \rightarrow K''_3 = 1.4584812 \dots$ as $N \rightarrow \infty$. (Source: [R2, Thm 10.1], base T3.)

The identity for K_3 reflects an image argument: the reflecting box with a corner target is equivalent to a periodic box of side $2N$ whose target is a block of $2 \times 2 \times 2$ sites. The second constant has a further form that is identified only numerically (Supplementary Section S5.3):

Numerical observation 4.9 (The two parts of K'_3). With $\xi_H = -\int_0^\infty [\theta_3(e^{-\pi u})^3 - 1 - u^{-3/2}] du = 2.8372974795$ and $\varkappa = -\int_0^\infty [\theta_4(e^{-t})^3 - 1] dt = 2.5193561521$, the constants of Theorem 4.8 satisfy $c_{\tau 3} = -\frac{6}{\pi} \xi_H$ and $c_{m\tau 3} = \frac{6}{\pi^2} \varkappa$ to within 3.7×10^{-14} and 2.8×10^{-14} , the precision to which ξ_H and $\varkappa$ were evaluated. These identities are not proved.

With $\mu_1 = \frac{2}{3} \sin^2(\pi/2N)$,

$$X = \mu_1 q T_N^{(3)} \approx \frac{\pi^2}{6} (K_3 N + K'_3) = 7.107 N - 6.39, \quad (28)$$

which exceeds the exact X by 0.058 at $N = 100$ and by 0.006 at $N = 1000$; for every $N \geq 2$ the difference $X - \frac{\pi^2}{6} (K_3 N + K'_3)$ lies between $-5.85/N - 159.0/N^2$ and $169.62/N^2$ (computer-assisted; [R2, Cors 11.1, 11.2]). In terms of the number of sites $n = N^d$, the laws $n \ln n$ for $d = 2$ and n for $d = 3$ have the form found by [Montroll \(1969\)](#) for a single trap on a periodic lattice.

Table 1: Coefficients of the expansion (24), with $\Xi = \varpi^4/\pi^2$ (digits truncated). C_2 , C_0 and C_{-2} are derived by hand (Supplementary Section S4); the closed forms marked † were obtained by computer algebra from the general coefficient formula and are confirmed numerically, not derived by hand. Last two columns: absolute difference between the closed form and (i) a Richardson extrapolation of exact 50-digit values of qT_N up to $N = 524288$, (ii) the solution of an independent linear system for the coefficients of twelve even powers, $N^2, \dots, N^{-20}$, from 70-digit values at $N = 2^j$, $6 \leq j \leq 17$, which uses no closed form.

	closed form		value	(i)	(ii)
C_2	$\frac{8}{\pi} [\gamma_E + \ln(4\sqrt{2}/\varpi) - \frac{1}{2} - \frac{\pi}{4}]$		0.154 637 787 818 ...	—	3×10^{-67}
C_0	$\Xi/9$		0.532 140 883 276 ...	6×10^{-46}	2×10^{-57}
C_{-2}	$-\pi\Xi(25\Xi + 648)/10800$		-1.069 558 947 686 ...	1×10^{-41}	2×10^{-48}
$C_{-4}^\dagger$	$\pi^2\Xi^2(1225\Xi + 12879)/1905120$		2.227 516 311 176 ...	3×10^{-33}	7×10^{-40}
$C_{-6}^\dagger$	$-\pi^3\Xi^2(79625\Xi^2 + 1215000\Xi + 960498)/435456000$		-14.055 169 191 762 ...	8×10^{-25}	4×10^{-32}
$C_{-8}^\dagger$	$\pi^4\Xi^3(7703465\Xi^2 + 92129400\Xi + 118943883)/63228211200$		124.706 087 396 014 ...	2×10^{-16}	8×10^{-25}

5 Exact modes and their dependence on dimension

This section reports the exact modes t^* for corner-to-corner transport (end to end for $d = 1$) and compares their growth with N with that of the means. Discrete-time values are for $q = 0.8$. In each of the 81 discrete-time runs used here (33, 31 and 17 sizes for $d = 1, 2, 3$) the maximiser is unique and certified to be global (Proposition 3.1), and all 81 coincide with the modes certified in exact integer arithmetic in Theorem 7.15. Continuous-time values are for unit jump rate and enter only through qt^* and t^*/T , which differ from their discrete-time counterparts by an offset of at most about three steps in t^* . Selected values are collected in Table 2, and Fig. 1 shows mean and mode against N ; the full tables, the compensated plots, the local exponents and the mode-to-mean ratio are in Supplementary Section S6.

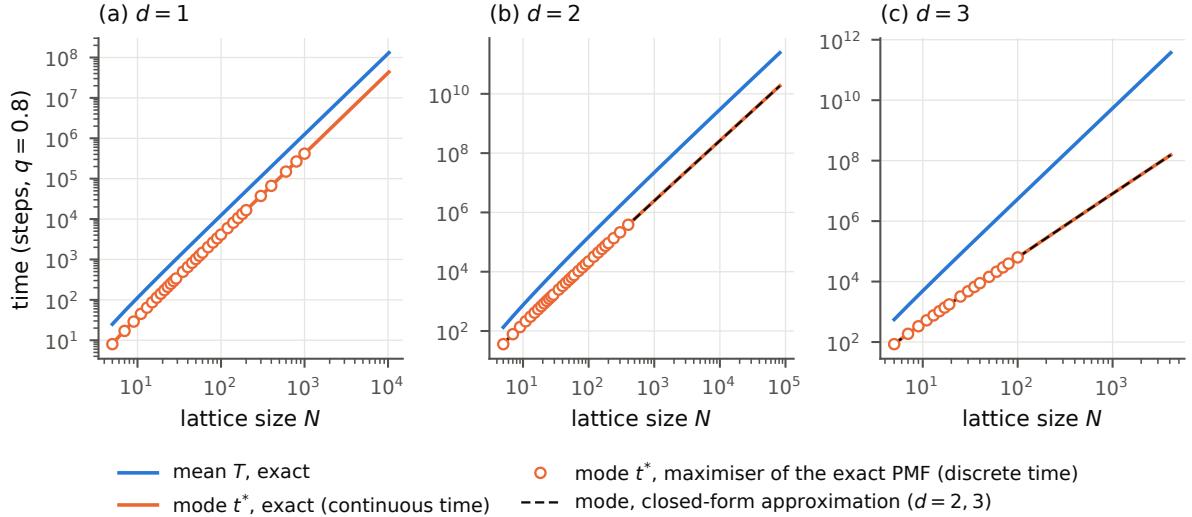


Figure 1: Mean first-passage time T and mode t^* against lattice size, corner to opposite corner (end to end for $d = 1$), in steps at $q = 0.8$. Lines: exact values in continuous time (unit rate, divided by q). Circles: maximiser of the exact discrete-time PMF. Dashed: closed-form approximation (34).

Table 2: Exact modes, corner to opposite corner. t^* : smallest maximiser of the exact discrete-time PMF at $q = 0.8$, in steps; unmarked rows are certified global maximisers (Proposition 3.1). Rows marked $\dagger$ are continuous-time modes at unit rate (Laplace inversion), for which qt^* stands for t_c^* . Rows marked $\ddagger$: local maximum of (29) nearest the continuum prediction, confirmed in 50-digit arithmetic; by Theorem 7.13(c) (computer-assisted) it is the global maximiser. Limit for $d = 1$: $2\xi^* = 0.3332843$ (31). qt^*/N^2 is given to four decimals and t^*/T to four significant digits (to seven decimals for $d = 1$, to show the approach to the limit).

d	N	t^* (steps)	qt^*/N^2	t^*/T
1	5	8	0.2560	0.3200000
1	50	1020	0.3264	0.3330612
1	100	4124	0.3299	0.3332525
1	200	16 580	0.3316	0.3332663
1	1000	416 188	0.3330	0.3332837
1	$10^5 \ddagger$	4 166 011 658	0.3333	0.3332843
1	$10^6 \ddagger$	416 604 915 263	0.3333	0.3332843
2	5	36	1.1520	0.2696
2	35	2493	1.6281	0.1768
2	100	22 110	1.7688	0.1489
2	200	92 116	1.8423	0.1350
2	401	383 013	1.9055	0.1236
2	1810 $\dagger$	–	2.0174	0.1048
2	20 480 $\dagger$	–	2.1523	0.08462
2	1 048 576 $\dagger$	–	2.3077	0.06509
2	16 777 216 $\dagger$	–	2.3905	0.05622
3	5	85	2.7200	0.1531
3	35	6644	4.3389	0.02945
3	40	8863	4.4315	0.02623
3	100	62 998	5.0398	0.01177
3	320 $\dagger$	–	5.7737	0.004188
3	1280 $\dagger$	–	6.6268	0.001199
3	4096 $\dagger$	–	7.3366	0.0004147

5.1 One dimension

In one dimension the whole distribution is available in closed form (proofs in Supplementary Section S6.1).

Proposition 5.1 (First-passage law of the chain, $d = 1$). *For the walk on $\{0, \dots, N-1\}$ started at the reflecting end 0 with target $N-1$, put $\omega_m = (2m-1)\pi/(2N-1)$ and $\lambda_m = 1 - q + q \cos \omega_m$, $m = 1, \dots, N-1$. Then $T = N(N-1)/q$ and, for all $t \geq 1$,*

$$f(t) = \frac{2q}{2N-1} \sum_{m=1}^{N-1} (-1)^{m+1} \cos \frac{\omega_m}{2} \sin \omega_m \lambda_m^{t-1}. \quad (29)$$

Put $\ell = N - \frac{1}{2}$, so that $\omega_m = (2m-1)\pi/(2\ell)$ exactly, and let

$$\Phi(\xi) = \pi \sum_{m \geq 1} (-1)^{m+1} (2m-1) \exp \left[-\frac{(2m-1)^2 \pi^2}{4} \xi \right], \quad \xi > 0, \quad (30)$$

the first-passage density of diffusion across an interval from a reflecting to an absorbing end, in units of the diffusion time (Redner, 2001); it has mean $\frac{1}{2}$ and variance $\frac{1}{6}$.

Proposition 5.2 (Mode-to-mean ratio of the chain). *(a) Φ is log-concave and real-analytic on $(0, \infty)$ and has a unique maximiser ξ^* , which lies in $(0.05, 0.5)$.*

(b) For the continuous-time walk on the chain, at any activity, $t_c^*/(qT) \rightarrow 2\xi^*$ as $N \rightarrow \infty$.

(c) For the discrete-time walk with $q \leq \frac{1}{2}$, $t^*/T \rightarrow 2\xi^*$ as $N \rightarrow \infty$.

The maximiser, computed in 30-digit arithmetic, is $\xi^* = 0.166642132747$, so that

$$2\xi^* = 0.3332842655, \quad (31)$$

which is the constant c of Theorem 7.13, enclosed there to within 10^{-70} . It lies below $\frac{1}{3}$ by 4.9×10^{-5} : the first term of the image expansion of Φ peaks at exactly $\xi = \frac{1}{6}$, the mode $(x_0 - x_a)^2/6$ of direct trajectories discussed by Godec and Metzler (2016) (its three-dimensional counterpart $(r - \rho)^2/(6D)$ is Eq. (3) of Grebenkov et al. (2018)), and the further images move the maximum. Proposition 5.2 does not cover the discrete walk at $q = 0.8$, where some λ_m are negative; Theorem 7.13 (computer-assisted) does, for every $q \in (0, 1]$. At $q = 0.8$ the exact modes are $t^* = 1020, 4124, 16\,580$ and $416\,188$ for $N = 50, 100, 200$ and 1000 , with $t^*/T = 0.33306, 0.33325, 0.33327$ and 0.3332837 , and the modes at $N = 10^5$ and 10^6 give 0.33328427 . The one-dimensional law has a fixed shape: in the continuum limit $\mathbb{P}(\mathcal{T} \leq t^*) = 0.1665$, the median is $0.7575T$, the coefficient of variation is $\sqrt{2/3} = 0.8165$ and the maximum of the density is $0.925/T$.

5.2 Two and three dimensions

Numerical observation 5.3 (Two dimensions). The compensated mode qt^*/N^2 equals 1.152, 1.628, 1.769, 1.842 and 1.906 at $N = 5, 35, 100, 200$ and 401 and, in continuous time, 2.017, 2.152, 2.308 and 2.390 at $N = 1810, 20\,480, 1\,048\,576$ and $16\,777\,216$; it increases, and t^*/T decreases, from each computed size to the next (31 sizes in discrete time, 43 in continuous time). The ratio t^*/T falls from 0.2696 at $N = 5$ to 0.0562 at $N = 16\,777\,216$.

The two-dimensional mode is therefore not proportional to N^2 : its prefactor changes by 60% over $5 \leq N \leq 200$ and keeps growing over the next five decades. The growth is slow: the local exponent $d \ln t^*/d \ln N$ falls from 2.27 near $N = 6$ to 2.06 near $N = 134$ and 2.013 near $N = 3 \times 10^6$, while that of the mean falls from 2.54 to 2.20 and 2.07. The leading law $qt^* \simeq (4/\pi^2)N^2[\ln \ln N + \ln 8\pi]$ of Section 6 is a theorem with an explicit remainder (Theorem 7.19), but the data do not display it: against $\ln \ln N$ the local slope of qt^*/N^2 decreases from 0.587 near $N = 6$ to 0.451 near $N = 1.2 \times 10^7$, still 11% above $4/\pi^2 = 0.405$, and the constant $\ln 8\pi = 3.22$ exceeds $\ln \ln N = 2.81$ at the largest size computed; the proved remainder decays only like $\ln \ln N / \ln N$. We therefore describe the two-dimensional mode by the closed form (35), and use ‘ $N^2 \ln \ln N$ ’ as the name of its leading asymptote.

Numerical observation 5.4 (Three dimensions). The compensated mode qt^*/N^2 equals 2.72, 4.43 and 5.04 at $N = 5, 40$ and 100 and, in continuous time, 5.77, 6.63 and 7.34 at $N = 320, 1280$ and 4096; it increases, and t^*/T decreases, from each computed size to the next (17 sizes in discrete time, 27 in continuous time). Against $\ln N$ its local slope is 0.959 near $N = 6$, 0.636 near $N = 134$ and 0.6095 near $N = 3200$. The ratio t^*/T is 0.1531, 0.0262 and 0.0118 at $N = 5, 40$ and 100, and 4.1×10^{-4} at $N = 4096$.

Over the computed range the three-dimensional mode thus grows as $N^2 \ln N$, not as N^3 : the measured slope 0.6095 is within 0.3% of the coefficient $6/\pi^2 = 0.6079$ of the proved law (Theorem 7.19), the local exponent of the mode falls to 2.085 near $N = 3200$ while that of the mean is 3.0003, and t^*/T falls like $\ln N / N$. The three-dimensional maximum is very flat: at $N = 40$ the two neighbours of the maximiser lie below the maximum by relative amounts of only 4×10^{-10} and 2×10^{-9} . Its position is resolved by exact computation (three time-stepping codes and the exact certificates agree), but it is sensitive to small errors in the PMF (Section 6.4).

Table 3: Mean, mode and their ratio, corner to opposite corner. Means: Theorems 4.1 and 4.5 ($d = 1, 2$) and Theorem 4.8 ($d = 3$). Modes: for $d = 1$ the continuum constant (31); for $d = 2, 3$ ‘ $\approx$ ’ marks the closed form (35) with $X = \mu_1 q T$ ((26), (28)), whose scaled error is bounded for every $N \geq 12$ ($d = 2$) and $N \geq 10$ ($d = 3$) (Theorem 7.18), and ‘ $\simeq$ ’ the leading laws (Theorem 7.19).

d	mean qT	mode qt^*	ratio t^*/T
1	$N(N-1)$	$0.33328(N - \frac{1}{2})^2$	$\rightarrow 0.3332843$
2	$\frac{8}{\pi}N^2 \ln N + C_2 N^2 + \dots$	$\simeq \frac{4}{\pi^2}N^2 [\ln \ln N + \ln 8\pi]$	$\approx \frac{\ln(4X)}{X+3} \simeq \frac{\ln(8\pi \ln N)}{2\pi \ln N}$
3	$K_3 N^3 + K'_3 N^2 + \dots$	$\simeq \frac{6}{\pi^2}N^2 [\ln N + 3.753]$	$\approx \frac{\ln(6X)}{X+5} \simeq 0.1407 \frac{\ln N + 3.753}{N}$

5.3 The separation between mean and mode grows with dimension

Table 3 summarises the three cases. At the common size $N = 100$ the mean exceeds the mode by the factors 3.0, 6.7 and 85 for $d = 1, 2, 3$; the factor is constant in one dimension, grows logarithmically in two and almost linearly in three. The reason, made quantitative in Section 6, is that the mode is tied to the time of relaxation across the box, of order N^2 in every dimension, whereas the mean is tied to the time needed to find the target. That the most probable first-passage time can lie far below the mean is known for diffusion in confined continuous domains (Godec and Metzler, 2016; Grebenkov et al., 2018); the lattice values and laws given here are exact counterparts for the lazy walk.

Scaling fits over short ranges of N cannot reveal these laws. The data are exact, so least-squares intervals measure misfit, not noise. Fitted on $10 \leq N \leq 200$, a free power law AN^a gives $a = 2.093 \pm 0.011$ for $d = 2$, and the exponent drifts with the window (2.019 ± 0.001 on $10^3 \leq N \leq 1.7 \times 10^7$); extrapolated to $N = 16\,777\,216$, the pure quadratic law is off by -31% , the free power law by $+125\%$, the form $N^2(A \ln \ln N + B)$ by $+3.8\%$ and the closed-form approximation (34), which has no fitted parameter, by -0.45% . In three dimensions a cubic law fitted on $10 \leq N \leq 100$ over-predicts the mode at $N = 4096$ by a factor 69, and the closed-form approximation errs by 6×10^{-7} (Supplementary Table S5).

6 Pole structure, a closed-form approximation of the mode, and the shape of the law

This section explains the results of Section 5; Section 7 states which parts of the explanation are proved. We work with the continuous-time walk at unit rate, whose density is $g(t) = \sum_j b_j e^{-\nu_j t}$ (6) and whose mean is qT ; the discrete-time walk has the same rates and residues. The propagator of the walk without target in the time domain is $p(\mathbf{x}, t | \mathbf{y}) = \sum_{\mathbf{k}} \phi_{\mathbf{k}}(\mathbf{x}) \phi_{\mathbf{k}}(\mathbf{y}) e^{-\mu_{\mathbf{k}} t}$.

6.1 Arrival and capture: a factorisation

Proposition 6.1 (Factorisation of the first-passage transform). *Let $\mathbf{o} \neq \mathbf{a}$, let $u(t) = n p(\mathbf{a}, t | \mathbf{o})$ be the occupation of the target site at time t in the walk without target, normalised by its stationary value $1/n$, and let $\mathcal{T}_{\mathbf{u}}$ be the first-passage time to $\mathbf{a}$ when the start is drawn uniformly from Ω (so that $\mathcal{T}_{\mathbf{u}} = 0$ with probability $1/n$), with transform $\hat{F}_{\mathbf{u}}(s) = \mathbb{E}[e^{-s\mathcal{T}_{\mathbf{u}}}]$. Then, for $s > 0$,*

$$\hat{F}_{\mathbf{u}}(s) = \frac{1}{n s \hat{P}(\mathbf{a}, s | \mathbf{a})}, \quad \hat{F}(s) = [s \hat{u}(s)] \hat{F}_{\mathbf{u}}(s), \quad (32)$$

that is, $g(t) = \int_{[0,t]} u'(t-\tau) g_u(d\tau)$. For the corner start and the opposite corner target,

$$u(t) = \left[1 + 2 \sum_{k=1}^{N-1} (-1)^k c_k^2 e^{-\varepsilon_k t} \right]^d, \quad \varepsilon_k = \frac{2}{d} s_k^2, \quad (33)$$

and for every fixed $x > 0$, $u(x/\mu_1) \rightarrow \theta_4(e^{-x})^d$ as $N \rightarrow \infty$, where $\theta_4(y) = 1 + 2 \sum_{k \geq 1} (-1)^k y^{k^2}$ is the Jacobi theta function of the nome y (DLMF, 2026, §20.2).

The proof (Supplementary Section S7) averages the renewal ratio over the start. The identity is the renewal equation written differently, and we do not claim it as new; its use is that it separates the two time scales of the problem. The factor u describes how the walker *arrives*: it knows nothing about the target and lives on the diffusive scale $1/\mu_1 \propto N^2$. The factor $\hat{F}_u$ describes *capture* from equilibrium, which for a reversible chain is close to exponential, with an error controlled by the ratio of the relaxation time to the mean hitting time (Aldous and Brown, 1992), here of order $1/X$. In one dimension $X = 2s_1^2 N(N-1) \rightarrow \pi^2/2$ is not large and the two scales coincide; in two and three dimensions X grows as $2\pi \ln N$ and as $7.107 N$, so the target is a weak sink on the diffusive scale.

6.2 Formal two-pole asymptotics

Formal result 6.2 (Closed-form approximation of the mode). Let μ be the smallest non-zero relaxation rate (unit rate) of the reflecting lattice whose eigenspace has non-zero weight at the target, $W = n \sum_{\mu_k=\mu} \phi_k(\mathbf{a})^2$ that weight and $A = -n \sum_{\mu_k=\mu} \phi_k(\mathbf{a}) \phi_k(\mathbf{o})$ the corresponding amplitude seen from the start, and let $X = \mu qT$. If $A > 0$ and $X \gg 1$, the mode of the continuous-time first-passage density at unit rate is

$$t_c^* \approx qT \frac{\ln(AX)}{X + W - 1}, \quad (34)$$

and the discrete-time mode at activity q is $t^* \approx t_c^*/q$. For corner-to-corner transport $\mu = \mu_1 = \frac{2}{d} \sin^2 \frac{\pi}{2N}$ and $A = W = 2d \cos^2 \frac{\pi}{2N}$; replacing $\cos^2$ by 1 gives the simplified form

$$\frac{t^*}{T} \approx \frac{\ln(2dX)}{X + 2d - 1}, \quad X = \mu_1 qT. \quad (35)$$

We write $t_{cf} := T \ln(2dX)/(X + 2d - 1)$ for the approximation of the discrete-time mode given by (35); in Section 7 and in the accuracy statements of the Introduction and of Section 9, “the closed form” means t_{cf} , whereas “the closed-form approximation” (34) keeps the factors A and W .

Derivation. Separate the uniform eigenvector (rate 0) and the eigenspace of μ :

$$n \hat{P}(\mathbf{a}, s | \mathbf{a}) = \frac{1}{s} + \frac{W}{s + \mu} + R(s), \quad \hat{u}(s) = n \hat{P}(\mathbf{a}, s | \mathbf{o}) = \frac{1}{s} - \frac{A}{s + \mu} + R_o(s), \quad (36)$$

where R and R_o collect the eigenvalues $\mu_k > \mu$ and are analytic for $\text{Re } s > -\mu'$, μ' being the next rate with weight at the target. The poles of $\hat{F}$ are the zeros $-\nu_j$ of the first function, with $0 < \nu_0 < \mu < \nu_1 < \mu'$ (Proposition 2.3), and $b_j = \hat{u}(-\nu_j)/n \partial_s \hat{P}(\mathbf{a}, s | \mathbf{a})|_{s=-\nu_j}$. The limit $s \rightarrow 0$ of $(1 - \hat{F})/s$ gives $qT = (W + A)/\mu + R(0) - R_o(0)$. Assume that $X \gg 1$, that R varies by a relative amount $O(1/X)$ on $[-\nu_1, 0]$ with $R(0) = qT(1 + O(1/X))$, and that $R_o = O(1/\mu)$ there. The two zero conditions then give

$$\nu_0 = \frac{1}{qT}(1 + O(X^{-1})), \quad \nu_1 = \mu \left(1 + \frac{W}{X} + O(X^{-2}) \right), \quad b_0 \approx \nu_0, \quad b_1 \approx -A \nu_0. \quad (37)$$

Keeping only $j = 0, 1$ in $g'(t) = -\sum_j b_j \nu_j e^{-\nu_j t}$ and solving $g' = 0$ (possible because $b_1 < 0$, that is, $A > 0$) gives $t_c^* = \ln(-b_1 \nu_1 / b_0 \nu_0) / (\nu_1 - \nu_0)$. To leading order,

$$t_c^* \approx \frac{\ln(AX)}{\mu}, \quad (38)$$

and inserting the first-order rate difference $\nu_1 - \nu_0 = \mu(X + W - 1)/X$ while keeping the argument of the logarithm at its leading value AX gives (34).

Status. This is a derivation, not a proof: it needs (i) uniform bounds on R and R' , (ii) a bound on the poles $j \geq 2$, and (iii) unimodality. For corner-to-corner transport in two and three dimensions Section 7 supplies all three by a route that does not expand R , and proves that $X\mu_1(t_c^* - qt_{cf})$ stays in a bounded window, a relative error $O(1/(X \ln X))$ for (35) (Theorem 7.18); the relations (37) are Theorem 7.5(c). In one dimension the assumption $X \gg 1$ fails and so does the formula (Refuted statement 7.14). Equation (34) is not a consistent expansion to order $1/X$; the omitted terms partly cancel (Supplementary Section S7.2).

6.3 Accuracy and leading laws

Table 4 compares the predictions with the exact continuous-time modes, on the computed grid of 41 sizes with $10 \leq N \leq 16\,777\,216$ for $d = 2$ and 25 sizes with $10 \leq N \leq 4096$ for $d = 3$. The leading-order form (38) errs by +19.9% at $N = 10$ and still by +2.4% at $N = 16\,777\,216$ for $d = 2$. The closed-form approximation (34) is within 0.972% of the exact mode for $d = 2$ (worst at $N = 113$) and within 0.289% for $d = 3$ (worst at $N = 10$; within 1.3×10^{-5} for $N \geq 100$); the simplified form (35) is within 0.970% and 0.085%. In two dimensions the error decreases slowly, to -0.45% at $N = 16\,777\,216$, where X is only 104.9. The errors of (35) at the 40 sizes with $N \geq 12$ ($d = 2$) and at all 25 sizes ($d = 3$) lie inside the windows of Theorem 7.18; for $d = 2$, $N = 10$ no window is proved. With the two-term mean of Theorem 4.5 and X from (26), so that no exact input is needed, the error of (35) stays within 0.975% for $d = 2$. The exact three-exponential truncation is within 5.35×10^{-5} ($d = 2$) and 7.1×10^{-6} ($d = 3$): the mode is controlled by very few poles.

With $1/\mu_1 \simeq 2dN^2/\pi^2$, $X \simeq 2\pi \ln N$ for $d = 2$ and $X \simeq \pi^2 K_3 N/6$ for $d = 3$, the leading-order form (38) becomes

$$d = 2 : \quad qt^* \simeq \frac{4}{\pi^2} N^2 [\ln \ln N + \ln 8\pi], \quad \frac{t^*}{T} \simeq \frac{\ln(8\pi \ln N)}{2\pi \ln N}, \quad (39)$$

$$d = 3 : \quad qt^* \simeq \frac{6}{\pi^2} N^2 [\ln N + \ln(\pi^2 K_3)] = 0.6079 N^2 (\ln N + 3.7528), \quad (40)$$

$$\frac{t^*}{T} \simeq 0.1407 \frac{\ln N + 3.753}{N}.$$

Both laws are theorems with explicit remainders (Theorems 7.19 and 7.20). In three dimensions the law is visible in the data: it exceeds the exact t_c^*/N^2 by 8.7% at $N = 10$, 0.81% at $N = 100$ and 0.02% at $N = 4096$. In two dimensions it is not: the constant $\ln 8\pi = 3.22$ exceeds $\ln \ln N$ for every $N < e^{8\pi} \approx 8 \times 10^{10}$, and the law exceeds the exact value by 19%, 8.9% and 2.3% at $N = 10$, 100 and 16 777 216. At every accessible size the two-dimensional mode should be described by (35), not by its leading term.

6.4 Shape of the law: the mode is not a typical time

Proposition 6.1 also describes the whole density. If g_u is replaced by the exponential law of rate ν_0 , then $g(t) \approx \nu_0 \int_0^t u'(\tau) e^{-\nu_0(t-\tau)} d\tau$, which is $\nu_0 u(t)$ for $t \ll qT$ and $\nu_0 e^{-\nu_0 t}$ for $t \gg 1/\mu_1$, and whose maximum is where the late-time arrival rate $u'(t) \approx A\mu e^{-\mu t}$ has dropped to the capture rate ν_0 , which is (38). Figure 2 shows both regimes for the exact densities: in the variable

Table 4: Signed relative error (in per cent, except the last column) of five predictions of the exact continuous-time mode, corner to corner, unit rate: the leading-order form (38); the closed-form approximation (34) with $A = W = 2d \cos^2(\pi/2N)$; its simplified form (35); and the exact two- and three-exponential truncations, that is, the maximiser of the two or three slowest exponentials with numerically exact rates and residues (last column: relative error, not per cent). No parameter is fitted.

	N	X	(38) (%)	(34) (%)	(35) (%)	two exp. (%)	three exp.
$d = 2$							
	10	14.740	+19.939	+0.2096	+0.2660	+0.6602	$+5.3 \times 10^{-5}$
	35	22.706	+12.231	-0.8354	-0.8222	+0.5134	$+5.5 \times 10^{-6}$
	113	30.083	+8.901	-0.9717	-0.9700	+0.3904	$+1.6 \times 10^{-6}$
	1280	45.335	+5.695	-0.8654	-0.8654	+0.2437	$+3.1 \times 10^{-7}$
	10 240	58.401	+4.361	-0.7384	-0.7384	+0.1791	$+1.2 \times 10^{-7}$
	1 048 576	87.485	+2.876	-0.5347	-0.5347	+0.1089	$+2.6 \times 10^{-8}$
	16 777 216	104.906	+2.392	-0.4545	-0.4545	+0.0871	$+1.4 \times 10^{-8}$
$d = 3$							
	10	64.168	+7.253	-0.2886	-0.0845	+0.1544	-7.1×10^{-6}
	35	242.186	+2.034	-0.0253	-0.0025	+0.0365	-3.1×10^{-7}
	100	704.235	+0.708	-0.0013	+0.0015	+0.0112	-2.9×10^{-8}
	320	2267.787	+0.221	+0.0005	+0.0008	+0.0031	-2.3×10^{-9}
	1280	9090.397	+0.055	+0.0002	+0.0002	+0.0007	-1.3×10^{-10}
	4096	29 103.352	+0.017	+0.0001	+0.0001	+0.0002	$+9.6 \times 10^{-11}$

Table 5: Shape statistics of the exact first-passage law, corner to corner, continuous time at unit rate: mode-to-mean ratio, probability of arrival by the modal time, median, survival at the mean, height of the density at the mode in units of $1/qT$ (equal to 1 for an exponential law), and the window, in units of t_c^* , in which the density exceeds 99% of its maximum. For $d = 1$ the start is the reflecting end.

d	N	t_c^*/qT	$\mathbb{P}(\mathcal{T} \leq t_c^*)$	median/ qT	$S(qT)$	$g(t_c^*) qT$	99% window
1	1000	0.3333	0.1665	0.7575	0.3708	0.9251	[0.89, 1.13]
2	35	0.1769	0.0981	0.7195	0.3683	0.9448	[0.87, 1.17]
2	200	0.1350	0.0780	0.7117	0.3681	0.9529	[0.85, 1.19]
2	163 840	0.0729	0.0456	0.7017	0.3679	0.9690	[0.82, 1.27]
3	40	0.0262	0.0174	0.6959	0.3679	0.9879	[0.77, 1.51]
3	100	0.0118	0.0082	0.6942	0.3679	0.9939	[0.72, 1.97]
3	1280	0.0012	0.0009	0.6932	0.3679	0.9992	[0.58, 9.47]

$x = \mu_1 t$ the product $qT g$ follows $\theta_4(e^{-x})^d$ up to the maximum, and in the variable $t/(qT)$ it follows $e^{-t/(qT)}$. There is no single-scale collapse.

Table 5 quantifies the consequence. For $d \geq 2$ the first-passage law to a distant point target is an exponential law preceded by a short dead time, not a peaked law with a long tail. Only 9.8% ($d = 2$, $N = 35$) and 1.7% ($d = 3$, $N = 40$) of the walkers have arrived by the modal time, and this fraction decreases with N ; the median is $0.72T$ and $0.70T$ and approaches $\ln 2 = 0.693$; the survival probability at the mean is 0.3683 and 0.3679 against $1/e = 0.3679$. The tail is exponential (geometric in discrete time) with rate ν_0 , not algebraic. The maximum is also flat, increasingly so with dimension and size: for $d = 3$ the density stays within 1% of its maximum from 0.58 to 9.5 modal times at $N = 1280$, and at $d = 3$, $N = 100$ a relative error of 1% in the density can displace the apparent maximum anywhere between 0.72 and 1.97 modal times. The mode is therefore a soft statistic: it marks the end of the dead time, it is not a typical arrival time, and the median is the appropriate typical value. Statements about waiting times must be made with S : the fraction of searches still unfinished is 36.8% at $t = T$, and 90% ($d = 2$,

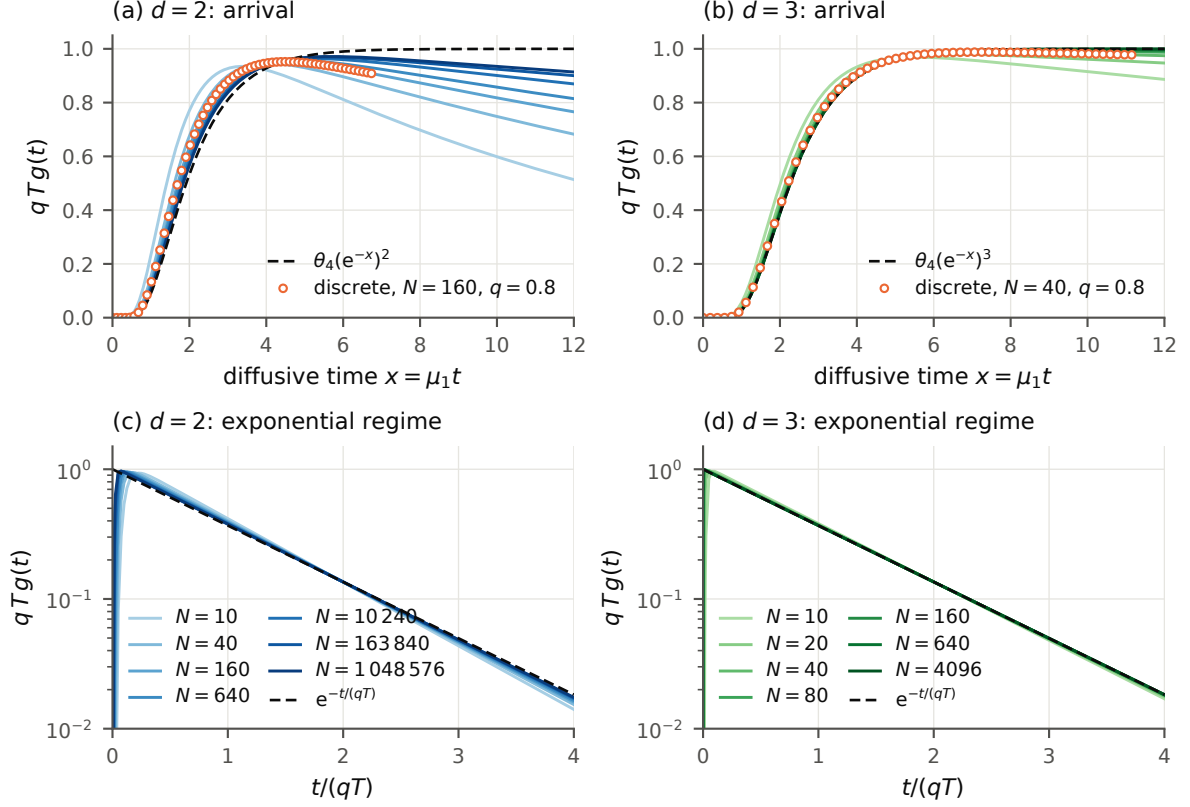


Figure 2: Exact first-passage densities, corner to corner, continuous time at unit rate (lines; sizes and colours as listed in panels (c) and (d)). (a,b) Against the diffusive time $x = \mu_1 t$, for $d = 2$ ($N = 10$ to $1\,048\,576$) and $d = 3$ ($N = 10$ to 4096); dashed: $\theta_4(e^{-x})^d$; open circles: exact discrete-time $Tf(t)$ at $q = 0.8$ for $N = 160$ ($d = 2$) and $N = 40$ ($d = 3$), plotted against $\mu_1 qt$. (c,d) The same densities against $t/(qT)$ on a logarithmic scale; dashed: $e^{-t/(qT)}$.

$N = 35$) or 98% ($d = 3$, $N = 40$) at $t = t^*$.

None of this structure is new in kind. Two time scales in confinement, and an exponential law rescaled by the mean when the target is hard to find, are established for continuous and scale-invariant processes (Bénichou et al., 2010; Godec and Metzler, 2016), and the rescaled laws fall into universality classes (Meyer et al., 2011); that the mean may then be unrepresentative, and the most probable time far below it, is the subject of Mattos et al. (2012), Godec and Metzler (2016) and Grebenkov et al. (2018); and the exponential approximation for reversible chains is classical (Aldous and Brown, 1992). In particular, Grebenkov et al. (2018, Eq. (31)) note that the median is close to $\ln 2$ times the mean when the survival probability is nearly exponential, which explains the limit $\ln 2$ above, and give the most probable time of the direct trajectories, $(r - \rho)^2/(6D)$ for a spherical target of radius ρ at distance r (Grebenkov et al., 2018, Eq. (3)), the continuum counterpart of the first-image term of Section 5. That estimate does not locate the mode of corner-to-corner transport for $d \geq 2$: there the mode is set by the relaxation of the box, $t^* \approx \ln(AX)/(q\mu)$ for large X by (34), a time of order $N^2 \ln(AX)$ whose logarithmic factor grows with N , whereas the direct-trajectory time is of order N^2 with no such factor. The relations (37) are the lattice analogue of the perturbation of eigenvalues by a small removed region (Ward and Keller, 1993), and uniform approximations of the density to a small target were given by Isaacson and Newby (2013), who obtain the mode by maximising the density numerically, without a closed formula. What is added here is the exact computation for the lattice walk, the dimension dependence of its mode, and the explicit closed-form approximation (34) and closed form (35) with their measured accuracy; we do not know whether these formulas

appear elsewhere and make no claim of priority.

6.5 Other placements and the activity

For a corner start and a centre target (C2M, odd N) the modes with $k_i = 1$ vanish at the target, so that $\mu = \frac{2}{d} \sin^2 \frac{\pi}{N} \approx 4\mu_1$, $W = 2d$ and $A = 2d \cos \frac{\pi}{N}$, and (34) is within 0.964% ($d = 2$, 24 sizes, $11 \leq N \leq 655\,361$) and 0.206% ($d = 3$, 15 sizes, $11 \leq N \leq 1281$) of the exact mode. For a centre start and a corner target (M2C) the formula does not apply: the slowest mode seen by the target vanishes at the start ($A = 0$), the residue b_1 is positive, and the maximum is set by the first arrivals, to which several poles contribute; the mode is nevertheless far below the mean, with exact discrete ratios 0.0352 ($d = 2$, $N = 201$) and 0.0046 ($d = 3$, $N = 61$) at $q = 0.8$. For exit from the centre through an absorbing boundary layer there is no small parameter, and t^*/T tends to a constant, 0.333284, 0.474907 and 0.560839 for $d = 1, 2, 3$. Details are in Supplementary Section S7.4.

By Proposition 2.2, qT and q times the continuous-time mode do not depend on q , and the discrete-time mode inherits this up to a few steps: for $d = 2$, $N = 101$ the exact argmax times q is 18 056.1, 18 056.0, 18 055.8 and 18 056 for $q = 0.3, 0.5, 0.9$ and 1, against $t_c^* = 18\,056.39$. Hence t^*/T does not depend on q beyond terms of order $1/T$.

7 What is proved about the first-passage law and its mode

Sections 5 and 6 rest on exact computation at the sizes run and on the formal argument of Section 6.2. This section states what is proved. Short proofs are in Supplementary Section S8; long proofs and the computer-assisted ones are in the companion notes.

7.1 Status of the statements, sources and checks

Companion notes. The complete proofs are contained in six notes (R0–R5, 32 to 51 pages each) in the folder `proofs/` of the repository (Zhouyi, 2026g), cited as [R0, ...], ..., [R5, ...] with their own statement numbers: R0, a dependency-ordered account of all results with a table of their status, the certificates and their trusted bases (Zhouyi, 2026a); R1, the walk on the segment (Zhouyi, 2026b); R2, the spectral structure of the first-passage law (Zhouyi, 2026c); R3, the mode of the corner-to-corner law in two and three dimensions (Zhouyi, 2026d); R4, unimodality (Zhouyi, 2026e); R5, exact certified computations (Zhouyi, 2026f). The notes number sites from 1 to N ; their Λ_1 , σ_j , r_j , $m(\mathbf{o})$, τ , C_3 , C'_3 are our μ_1 , ν_j , b_j/ν_j , qT , τ_u , K_3 , K'_3 .

Status words. *Theorem*, *Proposition*, *Lemma* and *Corollary* mean a complete proof. *Theorem (computer-assisted)* means that the proof reduces the statement, by lemmas proved by hand, to finitely many inequalities that a program verifies in exact integer or rational arithmetic or in rigorous ball arithmetic. Three trusted bases occur (labelled as in [R0, Section 9]; the bases T2 and T4 defined there do not enter the statements of this article): (T1) Python integers and fractions; (T3) ball arithmetic of the Arb library, through python-flint 0.9.0, together with the transcription of the formulas into the certification scripts; (T5) a C program of 366 lines with unsigned 64- and 128-bit integer arithmetic whose output is turned into verdicts with Python integers. For 2242 of the 3824 certificates of the C program, all beyond the ranges covered with Python integers, that program, run twice with identical checksums, is the only witness. Every statement names its base; the certificates and the programs that check them are in the repository. No program has been verified in a proof assistant, and floating-point arithmetic is used only to find candidates, never to decide. *Formal result* means a derivation by formal asymptotics, *Numerical observation* a statement established by exact computation at the sizes stated only, and *Refuted statement* a natural conjecture, suggested by the one-dimensional case or by numerical evidence, that is false.

Verification and reproducibility. The proof notes supply the analytical reductions and identify the arithmetic certificates supporting each computer-assisted statement. The scripts in the repository check the one-dimensional constants, the spectral bounds, the Taylor-model inequalities for unimodality, and the finite-range mode certificates. The discrete-time window for every $q < 1$ is verified under its stated size condition by regenerating the ball-arithmetic certificate and checking the complete block coverage. The commands and their coverage are documented in `submission/SCIENTIFIC-CHECKS.md`; the detailed derivations remain in the companion notes. The trusted bases and the parameter restrictions stated above are unchanged.

7.2 The first-passage law: structure and signs

Here the rates are indexed by all zeros of Proposition 2.3(i), $\nu_j := \nu_{(j)}$, with $b_j = 0$ when the pole is absent from the law started at $\mathbf{o}$. We write $r_j := b_j/\nu_j$ for the weights of the survival function, $S(t) = \sum_j r_j(1 - q\nu_j)^t$, so that $\sum_j r_j = 1$, and $D = |\mathbf{o} - \mathbf{a}|_1$ for the graph distance.

Theorem 7.1 (Signs of the residues). *For every symmetric chain of Section 2 (any N , d , start and target):*

- (a) $\sum_j r_j \nu_j^i = 0$ for $1 \leq i \leq D - 1$ and $(-1)^{D-1} \sum_j r_j \nu_j^D > 0$; hence the sequence (r_j) has at least $D - 1$ changes of sign. For the chain started at its reflecting end, the number of rates is $D = N - 1$ and the signs alternate, $\text{sign } r_j = (-1)^j$.
- (b) From the uniform start the law is, apart from the atom $1/n$ at $t = 0$, a mixture of exponential laws with positive weights (of geometric laws in discrete time, for $q \leq \frac{1}{2}$).
- (c) From the quasi-stationary distribution the first-passage time is exactly exponential with rate ν_0 (geometric in discrete time).

(Source: [R2, Thm 5.1, Props 5.4, 5.5], [R4, Thm 3.2]; (b) and (c) are classical for reversible chains (Aldous and Fill, 2002).)

Theorem 7.2 (A single distant start is never a mixture). *If $D \geq 2$, there is no positive measure m on $[0, \infty)$ with $g(t) = \int e^{-\sigma t} m(d\sigma)$ for all $t > 0$, and none on $[0, 1)$ with $f(t) = \int \lambda^{t-1} m(d\lambda)$ for all $t \geq 1$; in particular some residue b_j is negative.* (Source: [R4, Thm 4.1(d)].)

Theorem 7.3 (Signs of the corner-to-corner residues). *For the corner-to-corner pair: (a) if $d \geq 2$, then $r_0 > 0 > r_1$ for every $N \geq 2$; (b) if $d = 1$ or $N = 2$, the number of rates is $D = d(N - 1)$ and $\text{sign } r_j = (-1)^j$; (c) if $d \geq 2$ and $N \geq 3$, the residues do not alternate in sign: there are three consecutive residues whose signs form a monotone sequence.* (Source: [R2, Cor 5.7, Prop 5.9, Thm 5.10].)

Refuted statement 7.4. “The residues of the corner-to-corner law alternate in sign in every dimension, as they do in one dimension.” False for every $d \geq 2$ and $N \geq 3$ by Theorem 7.3(c). The smallest example is $d = 2$, $N = 3$, where the five residues have the signs $+, -, -, +, -$ (certified in ball arithmetic). (Source: [R2, Section 5.2, Prop 14.1].)

Descartes’ rule bounds the number of zeros of g' by the number of sign changes of the residues, which is at least $D - 1$; for $d \geq 2$ and $N \geq 3$ it cannot prove unimodality, and the proofs below use log-concavity instead. Let $\tau_u = q \mathbb{E} T_u$ be the unit-rate mean first-passage time from the uniform start (Proposition 6.1) and $\bar{X} = \mu_1 \tau_u$.

Theorem 7.5 (The two slowest poles). *(a) For every target, $\tau_u \leq 1/\nu_0 \leq \tau_u + 1/\mu_{(1)}$.*

- (b) *For the corner-to-corner pair, every $d \geq 1$ and $N \geq 2$ with $(d, N) \neq (1, 2)$: $r_0 > 1$ and $\nu_0 qT > 1$.*

(c) For the corner-to-corner pair in $d = 2, 3$, as $N \rightarrow \infty$,

$$\begin{aligned} \nu_0 qT &= 1 + \frac{\mu_\infty}{\bar{X}} + o(\bar{X}^{-1}), & r_0 &= 1 + \frac{\mu_\infty}{\bar{X}} + o(\bar{X}^{-1}), \\ \frac{\nu_1 - \nu_0}{\mu_1} &= 1 + \frac{2d-1}{\bar{X}} + O(\bar{X}^{-2}), & \frac{b_1}{b_0} &\rightarrow -2d, \end{aligned}$$

where $\mu_\infty := \lim_{N \rightarrow \infty} (X - \bar{X})$, a limit that exists, with $\mu_\infty = \pi \ln 2$ for $d = 2$ and $\mu_\infty = \frac{\pi^2}{6} c_{m\tau 3} = 2.51935 \dots$ for $d = 3$ ($c_{m\tau 3}$ of Theorem 4.8; this value agrees numerically with $\varkappa$ of Observation 4.9); moreover $\nu_1/\mu_1 = 1 + \delta$ with $W/\delta = \bar{X} - (2d + 1 - Z'_d) + o(1)$, where $Z'_d = \sum_{\mathbf{k} \in \mathbb{Z}^d, |\mathbf{k}|^2 \geq 2} (|\mathbf{k}|^2 (|\mathbf{k}|^2 - 1))^{-1}$ (Euclidean norm), $Z'_2 = 3.2259 \dots$ and $Z'_3 = 14.7022 \dots$

(Source: [R2, Thms 6.2, 7.9, 11.3, Lemma 7.6].)

Part (c) contains the relations (37) of the formal argument; they do not by themselves locate the mode [R2, Section 14, item (G1)], and Section 7.5 proves the mode laws by a different route. All poles beyond the second are controlled as well: for $d \geq 2$, $\bar{X} > 1$ and $t \geq 1/\mu_1$, the unit-rate density differs from its two slowest exponential terms by at most an explicit constant times $\bar{X}^{-1} \mu_1 e^{-2\mu_1 t}$ [R2, Thm 12.1, Cor 12.2] (Supplementary Section S8.6). Near the mode, where $e^{-\mu_1 t}$ is of order $1/\bar{X}$, this is a relative error $O(\bar{X}^{-2})$; at $t = 1/\mu_1$ it is of the same order as the two slowest terms themselves, so that the bound is not a small relative error on the whole range $t \geq 1/\mu_1$.

7.3 Unimodality

A PMF f is *unimodal* if for some t_0 it is non-decreasing for $t \leq t_0$ and non-increasing for $t \geq t_0$ (ties are allowed), and *strictly unimodal* if, after its initial zeros, it is strictly increasing up to t_0 and strictly decreasing after t_0 , so that its maximiser is unique [R5, Def 3]; a continuous-time density is unimodal with a unique mode if g' changes sign exactly once, from positive to negative. For example, at $d = 1$, $N = 4$, $q = \frac{4}{5}$ the PMF satisfies $0 = f(2) < f(3) = f(4) < f(5)$: it is unimodal but not strictly unimodal. Let $q_N^\square := 1/(1 + \cos \frac{\pi}{N})$; it exceeds $\frac{1}{2}$, equals 1 for $N = 2$ and decreases to $\frac{1}{2}$. For $q \leq q_N^\square$ all eigenvalues of P are non-negative.

Theorem 7.6 (The three point-target placements). *Let $d \geq 1$, $N \geq 2$, and let the pair be corner to corner, corner to centre or centre to corner (the last two for odd N), or the image of one of these under a symmetry of the lattice.*

(i) *For every rate the continuous-time density is unimodal with a unique mode. For the corner-to-corner pair with $(d, N) \neq (1, 2)$, g' has exactly one zero in $(0, \infty)$; it is positive before and negative after it*

(ii) *For every $q \in (0, q_N^\square]$ the PMF is unimodal.*

(Source: [R4, Thm 4.12], [R3, Thm 1.11]; for $d = 1$ the unimodality of (i) is classical (Keilson, 1971).)

The proof (Supplementary Section S8) is by hand. It writes the law as the arrival time of Proposition 6.1 smoothed by the escape kernel, uses that the one-dimensional factors of the arrival time are birth-and-death passage times with log-concave laws, and that log-concavity survives the (binomial) convolutions that build the lattice from its coordinates.

Theorem 7.7 (The value $q = \frac{4}{5}$; computer-assisted). *Let $q \in (0, \frac{4}{5}]$. The corner-to-corner PMF is unimodal for every $N \geq 2$ and $d \geq 1$, and the corner-to-centre and centre-to-corner PMFs are unimodal for every odd $N \geq 3$ and $d \geq 2$. The value $\frac{4}{5}$ is sharp (Proposition 7.8), and $d \geq 2$ cannot be dropped for the centre geometries: in $d = 1$ the centre-to-corner PMF at $q = \frac{4}{5}$ is unimodal if and only if $N \notin \{5, 7, 9, 11\}$. (Source: [R4, Props 9.1, 9.12, Thm 9.13], base T3 with T1 cross-checks.)*

The proof shows that the increments of the one-dimensional free propagator are log-concave for every $q \leq \frac{4}{5}$ and every $N \geq 9$ (centre geometries: every odd $N \geq 25$): beyond an explicit time by hand, below it in ball arithmetic for $N \leq 1001$ and, for $N \geq 992$, through Taylor models with Cauchy remainder bounds on 26 parameter boxes; a binomial-convolution identity carries this to every dimension, and exact cone certificates cover the finitely many smaller lattices [R4, Section 9].

Proposition 7.8 (Failure near $q = 1$ on the chain). *Let $d = 1$, start 0, target $N - 1$, $N \geq 4$. Then $f(N + 1) > f(N)$ for every $q \in (0, 1]$, and $f(N) < f(N - 1)$ if and only if $q > \frac{2N-4}{2N-3}$; for such q the PMF is not unimodal, in particular at $q = 1$. For $N \in \{2, 3\}$ it is unimodal for every q , and for $N = 4$ if and only if $q \leq \frac{4}{5}$. (Source: [R1, Prop 3.9], [R4, Prop 4.4], [R5, Prop 32].)*

Theorem 7.9 (Finite ranges and $q = 1$; computer-assisted). *For the corner-to-corner pair: (a) for $d = 2$, $2 \leq N \leq 64$ and every $q \leq \frac{499}{500}$ the PMF is unimodal; (b) for $d = 3$, $2 \leq N \leq 30$, $d = 4$, $N \leq 8$, and $d = 5$, $N \leq 5$, it is unimodal for every $q \in (0, 1]$, and for $d = 3$, $q = 1$, $2 \leq N \leq 60$ it is strictly unimodal; (c) for $d = 2$, $q = 1$, $2 \leq N \leq 160$ and $N \in \{180, 200\}$, it has two strict local maxima exactly for $N \in \{9, 11, 13, 15, 17, 18, 21, 24, 26, 38, 51, 61, 86, 92, 136, 149\}$ (for $N \leq 64$: one dip of one step next to the maximum) and is strictly unimodal otherwise, except $f(2) = f(3)$ at $N = 2$. (Source: [R4, Thm 7.1], [R5, Thms 13, 40], bases T1 and T5.)*

Theorem 7.10 (Bimodal pairs; computer-assisted). *(a) $d = 3$, $N = 3$, start $(2, 1, 1)$, target $(0, 1, 1)$ (centres of opposite faces): the unit-rate density satisfies $g(\frac{21}{10}) > g(3) < g(13)$, and at $q = \frac{1}{2}$, $f(2) = \frac{1}{144} > f(6) = \frac{6577}{995328} < f(25)$. (b) $d = 2$, $N = 8$, start $(5, 5)$, target $(2, 3)$: $g(\frac{21}{2}) > g(\frac{49}{2}) < g(\frac{59}{2})$, and at $q = \frac{1}{2}$, $f(19) > f(46) < f(63)$. In both cases neither the density at any rate nor the PMF at any q is unimodal. (Source: [R4, Thm 8.1], base T1.)*

Refuted statement 7.11. “The point-target first-passage law is always unimodal.” False in discrete time near $q = 1$ (Proposition 7.8, Theorem 7.9(c)), for starts away from the reflecting end of the chain when q is large [R1, Prop 8.8], and for other pairs in two and three dimensions even in continuous time (Theorem 7.10). (Source: [R4, Section 8], [R5, Section 9].)

Conjecture 7.12. *(a) For the corner-to-corner pair with $d \geq 2$, the law is log-concave in continuous time and, for $q \leq \frac{1}{2}$, in discrete time. (b) For $d \geq 3$ the corner-to-corner PMF is unimodal for every $q \leq 1$ and every N . (c) For $d = 2$ the largest q up to which the corner-to-corner PMF is unimodal for every N is the threshold of $N = 9$, which lies in $[0.9986, 0.9987)$ (certified). (Source: [R4, Conjs 10.1, 10.2].)*

7.4 One dimension

On the chain, start 0, target $N - 1$, $\ell = N - \frac{1}{2}$ and $T = N(N - 1)/q$. The constant $2\xi^* = 0.3332842655$ of (31) is written c here: it is the maximiser of the limit density $\Phi(u/2)/2$ in the time unit ℓ^2 .

Theorem 7.13 (The mode on the chain; computer-assisted). *(a) the constant is*

$$c = 0.33328\,42654\,94947\,89874\,85269\,16543\,44242\,10970\,36559\,07113\dots,$$

with every digit shown certified (the enclosure has radius 10^{-70}); with $G(u) = \frac{1}{2}\Phi(\frac{u}{2})$ the limit density in the time unit ℓ^2 , two further constants are

$$\begin{aligned}\alpha &:= \frac{3}{4} + \frac{c}{6} \frac{G'''(c)}{G''(c)} = 0.08627\,37793\,78239\,01718\,17\dots, \\ \beta &:= -\frac{c}{2} \frac{G'''(c)}{G''(c)} = 1.99117\,86618\,65282\,94845\,46\dots;\end{aligned}$$

- (b) in continuous time, for every $N \geq 3$, the unit-rate mode satisfies $|t_c^*/\ell^2 - c + \alpha\ell^{-2}| < 1.78\ell^{-4}$;
- (c) in discrete time with $q \leq \frac{4}{5}$ and $N \geq 3$ (also $q \leq \frac{17}{20}$ with $N \geq 7$, and $q \leq \frac{9}{10}$ with $N \geq 10$) the PMF is unimodal and every mode satisfies

$$\tau_\ell - E < t^* < \tau_\ell + 1 + E, \quad \tau_\ell = \frac{c\ell^2 - \alpha}{q} + 1 - \beta, \quad E = \frac{1.78}{q\ell^2}$$

($E = 1.68/(q\ell^2)$ for $N \geq 41$ and $q \leq \frac{4}{5}$); if $\text{dist}(\tau_\ell, \mathbb{Z}) \geq E$ the mode is unique and equals $\lceil \tau_\ell \rceil$;

- (d) for every fixed $q \in (0, 1]$, $t^*/T \rightarrow c$ as $N \rightarrow \infty$; the error is $O(N^{-2})$ for $q < 1$, whereas for $q = 1$ the mode has the parity of $N - 1$ and $\ell(c - t^*/\ell^2) \rightarrow b = 0.33250858299642786\dots$, so that the convergence is only of order N^{-1} .

(Source: [R1, Thms 4.6, 5.7, 6.1, 6.3, 7.4, 7.7, Cors 5.8, 6.4, 7.8], [R5, Thm 23], base T3 (with T1 for $q = 1$); α and β are the constants κ and ρ of [R1, Thm 4.6].)

The windows rest on an expansion of the lattice density and of its increments with remainders enclosed in ball arithmetic uniformly in $\ell \geq \ell_0$, on 2400 intervals of q in (c), and on a check of every smaller N ; part (d) for $\frac{4}{5} < q < 1$ uses global bounds without unimodality [R1, Sections 5–7]; the next coefficient of (b) is derived in the same note. The theorem extends Proposition 5.2(c) to every q and decides the value of the mode: for $q = 0.8$ the offset in (c) is $\alpha/q - 1 + \beta = 1.09902\dots$. For the 33 runs of the chain at $q = 0.8$ and the sizes $N = 10\,240$, 10^5 and 10^6 , every computed mode lies in the window of (c); in 34 of the 36 cases $\text{dist}(\tau_\ell, \mathbb{Z}) \geq E$, and there $\lceil \tau_\ell \rceil$ is the computed mode.

Refuted statement 7.14 (The closed form in one dimension). “ $t^* \approx T \ln(2dX)/(X + 2d - 1)$ in every dimension.” False for $d = 1$: there $X \rightarrow \pi^2/2$, so the closed form gives $t^*/T \rightarrow \ln(\pi^2)/(1 + \pi^2/2) = 0.385768\dots$ instead of c , an overestimate by a factor tending to $1.15747\dots$. On finite ranges (computer-assisted, base T1 with T3): $0.1577t^* \leq T \ln(2X)/(X + 1) - t^* \leq 0.2251t^*$ for $q = \frac{4}{5}$, $10 \leq N \leq 1500$ and $N = 2000$, and the same with 0.1579 and 0.2282 for $q = \frac{1}{2}$, $10 \leq N \leq 1000$. (Source: [R1, Rem 7.9], [R5, Thms 21, 43].)

Theorem 7.15 (Certified modes; computer-assisted). *For the corner-to-corner pair the PMF is strictly unimodal with a certified integer mode in each of the following cases, apart from three exact ties in one dimension: $d = 1$: $q = \frac{4}{5}$, $2 \leq N \leq 1500$ and $N = 2000$; $q = \frac{1}{2}$, $2 \leq N \leq 1000$; $d = 2$: $q = \frac{4}{5}$, $2 \leq N \leq 301$ and $N \in \{350, 401, 450, 500, 600\}$; $q = \frac{1}{2}$, $2 \leq N \leq 200$; $d = 3$: $q = \frac{4}{5}$, $2 \leq N \leq 120$ and $N \in \{130, 140, 150, 160\}$; $q = \frac{1}{2}$, $2 \leq N \leq 80$. The ties are $(q, N) = (\frac{4}{5}, 4)$, where $f(3) = f(4) < f(5)$ and the mode is 5, $(\frac{1}{2}, 3)$, with maxima at $t = 3, 4$, and $(\frac{1}{2}, 4)$, with maxima at $t = 7, 8$. For example $t^* = 872\,259$ for $d = 2$, $N = 600$ and $t^* = 170\,871$ for $d = 3$, $N = 160$, both at $q = \frac{4}{5}$. (Source: [R5, Thms 12, 13, 30, 39, 40, 45, Cors 18, 42], bases T1 and T5.)*

Each certificate steps the walk exactly up to a time beyond which every occupation probability decreases, which forces the PMF to decrease (Lemma S8.1). Every corner-to-corner discrete-time mode of Sections 3–6 at $q \in \{0.5, 0.8, 1\}$ lies in a certified range: the 81 modes at $q = 0.8$ and the eight at $q = 0.5$ (this theorem), and the eight at $q = 1$ (Theorem 7.9 for $d = 2, 3$; for the chain, where the PMF at $q = 1$ is not unimodal, [R5, Thms 13, 40] certify the global maximiser for $N \leq 400$); all 97 agree with the certified modes. The 16 corner-to-corner runs at $q = 0.3$ and 0.9 (two of which are quoted in Section 6.5) lie in no certified range; they rest on the a-priori round-off bound of Section 3.

Table 6: Constants of the windows of Theorems 7.18 and 7.20: for every $N \geq N_0$, $-K^- < X \cdot q\mu_1(t^* - t_{\text{cf}})$ and $X \cdot q\mu_1(t^* - t_{\text{cf}} - 2) < K^+$ in discrete time (without the -2 in continuous time, with t^* replaced by t_c^*/q). A negative K^- means that the mode lies above the closed form. Selected rows; the full tables are in [R3, Thms 5.1, 7.8, 7.14, 7.17].

case	d	$(N_0; K^-, K^+)$			
continuous time	2	(12; 7.653, 10.280)	(50; 2.058, 5.449)	(1000; -0.116, 4.641)	(10^6 ; -1.382, 4.641)
	3	(10; 3.120, 7.120)	(50; 1.160, 2.149)	(1000; 0.862, 1.465)	(10^6 ; 0.828, 1.427)
$q \leq \frac{1}{2}$	2	(12; 8.546, 10.392)	(50; 2.099, 5.462)	(1000; -0.116, 4.641)	
	3	(10; 5.293, 7.513)	(30; 1.962, 2.829)	(1000; 0.898, 1.465)	
$q \leq 0.8$	2	(37; 2.850, 6.613)	(100; 1.448, 5.560)	(1000; -0.016, 5.135)	
	3	(13; 5.189, 6.634)	(50; 1.851, 2.973)	(1000; 1.021, 2.141)	
every $q < 1$	2	(59; 2.082, 6.061)	(100; 1.508, 5.594)	(1000; -0.009, 5.135)	
	3	(12; 6.379, 7.183)	(100; 1.675, 2.495)	(1000; 1.036, 2.142)	

7.5 Two and three dimensions: the closed form

Throughout this subsection the pair is corner to corner and $d \in \{2, 3\}$. Let

$$t_{\text{cf}} := T \frac{\ln(2dX)}{X + 2d - 1}, \quad X = \mu_1 qT,$$

which is the closed form (35). At activity q the natural time variable is $q\mu_1 t$.

Proposition 7.16 (Arrival and capture). *The function u of (33) is the distribution function of a random time U : the largest of d independent sums of $N - 1$ independent exponential variables with rates $\varepsilon_1, \dots, \varepsilon_{N-1}$. Hence, at unit rate, $\mathcal{T} \stackrel{d}{=} U + \mathcal{T}_u$ with U and $\mathcal{T}_u$ independent, and $X = \bar{X} + \mu_1 \mathbb{E}U$. The densities of U and of its one-dimensional factors are log-concave. (Source: [R3, Lemmas 1.2–1.4, 1.7, Prop 1.5].)*

Theorem 7.17 (An exact identity and the window principle). *For $N \geq 3$ ($N \geq 4$ if $d = 3$), g' is the sum of a two-exponential term built from the two slowest poles and of three correction terms, each with a definite sign and explicit two-sided exponential bounds in terms of the pole data. Hence there are explicit functions $\Upsilon_- \leq g' \leq \Upsilon_+$, and by Theorem 7.6(i) $\Upsilon_-(t) > 0$ implies $t < t_c^*$ and $\Upsilon_+(t) < 0$ implies $t > t_c^*$. (Source: [R3, Prop 2.1, Lemma 2.2, Thm 2.3].)*

This is the rigorous form of the two-pole argument of Section 6.2. The pole data are bounded by explicit functions of $1/\bar{X}$ and of one lattice sum, and the lattice sums, X and $\mathbb{E}U$ by explicit functions of N for $N \geq 10$ (for example $2\pi \ln N - 0.24 \leq X \leq 2\pi \ln N + 0.60$ for $d = 2$; base T3) [R3, Lemmas 3.1–3.5, Props 4.5, 4.6, Cor 4.7]. Inserting these bounds at $\mu_1 t_c^* = q\mu_1 t_{\text{cf}} \pm c_{\pm}/X$ turns the sign conditions into finitely many inequalities on parameter blocks: 220 blocks for $d = 2$ and 389 for $d = 3$, single sizes for small N , geometric blocks up to $N = 10^{60}$ and 10^9 , and one unbounded block each.

Theorem 7.18 (The closed form in continuous time; computer-assisted). *For every $N \geq N_0$,*

$$-K_d^-(N_0) < X\mu_1(t_c^* - qt_{\text{cf}}) < K_d^+(N_0),$$

with the constants of Table 6. Consequently $|t_c^ - qt_{\text{cf}}|/(qt_{\text{cf}}) \leq K_d(1 + \frac{2d-1}{X})/(X \ln(2dX))$ with $K_d = \max(K_d^-, K_d^+)$: the relative error of the closed form is $O(1/(\ln N \ln \ln N))$ for $d = 2$ and $O(1/(N \ln N))$ for $d = 3$. (Source: [R3, Thm 5.1], base T3.)*

Theorem 7.19 (The leading laws; computer-assisted). *For every $N \geq 20$ the unit-rate mode satisfies*

$$d = 2 : \quad t_c^* = \frac{4N^2}{\pi^2} [\ln \ln N + \ln(8\pi) + \Delta_N](1 + \epsilon_N), \quad -\frac{4.6 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \leq \Delta_N < 0,$$

$$d = 3: \quad t_c^* = \frac{6N^2}{\pi^2} [\ln N + \ln(\pi^2 K_3) + \Delta_N] (1 + \epsilon_N), \quad -\frac{9.2 + 5 \ln(\pi^2 K_3 N)}{\frac{\pi^2}{6} K_3 N - 7.63} \leq \Delta_N < 0,$$

with $0 \leq \epsilon_N \leq 0.83/N^2$ and $\ln(\pi^2 K_3) \in [3.752820, 3.752823]$. In particular $\frac{4}{\pi^2} N^2 (\ln \ln N + 2.27) \leq t_c^* \leq \frac{4}{\pi^2} N^2 (\ln \ln N + 3.2242) (1 + \frac{0.83}{N^2})$ for $d = 2$ and $\frac{6}{\pi^2} N^2 (\ln N + 3.43) \leq t_c^* \leq \frac{6}{\pi^2} N^2 (\ln N + 3.7529) (1 + \frac{0.83}{N^2})$ for $d = 3$, and $t_c^*/qT \rightarrow 0$ like $\ln \ln N / (2\pi \ln N)$ and $6 \ln N / (\pi^2 K_3 N)$. (Source: [R3, Thms 6.1–6.3, Cor 6.4], base T3.)

These are the laws (39) and (40) with proved remainders. In two dimensions the remainder Δ_N decays like $\ln \ln N / \ln N$, which is why the computed sizes do not display the leading law.

Theorem 7.20 (Discrete time; computer-assisted). *Every maximiser of the PMF satisfies*

$$t_{\text{cf}} - \frac{K^-}{X q \mu_1} < t^* < t_{\text{cf}} + \frac{K^+}{X q \mu_1} + 2 \quad (N \geq N_0),$$

with constants that do not depend on q (selected values in Table 6), in each of the following cases: (a) $q \leq \frac{1}{2}$; (b) $q \leq q_{\max}$ for $q_{\max} \in \{0.8, 0.9, 0.95, 0.99\}$; (c) every $q < 1$, provided $N \geq \max(N_0, N_1(q))$, where $N_0 \geq 59$ ($d = 2$), $N_0 \geq 12$ ($d = 3$) and $N_1(q) = \min\{N \geq 4 : \sin \frac{\pi}{N} \leq \frac{1-q}{q}\}$, which is about $\pi q / (1 - q)$ as $q \rightarrow 1$. Consequently the laws of Theorem 7.19 hold for t^* at activity q (with t_c^* replaced by qt^* , the lower numerators 4.6 and 9.2 replaced by 4.72 and 10.30 in case (a), by 1.72 and 9.28 in case (b) for $N \geq 100$ ($N \geq 312$ if $q > 0.95$), and by 2.33 and 14.01 in case (c), and with 2 added to the upper bounds). (Source: [R3, Thms 7.7–7.9, 7.14, 7.15, 7.17, Cor 7.18], base T3.)

Case (a) combines certified signs of the increments at two times with unimodality (Theorem 7.6(ii)); cases (b) and (c) use no unimodality: under $N \geq N_1(q)$ a pairing lemma for the one-dimensional factors controls the terms with negative multipliers $1 - qv_j$ [R3, Thm 7.13]. The certificate of case (c) covers 171 blocks for $d = 2$ and 387 for $d = 3$, including the unbounded final block. Each block verifies the inequalities required by the theorem in ball arithmetic; the complete certificates can be regenerated using the commands in `reproduce.md`.

Theorem 7.21 (Accuracy on finite ranges; computer-assisted). *With the certified modes of Theorem 7.15 and rational enclosures of T (Lemma S8.2): (a) $d = 2$: $|t^* - t_{\text{cf}}| \leq 0.00970 t^*$ for $q = \frac{4}{5}$, $10 \leq N \leq 301$ and $N \in \{350, 401, 450, 500, 600\}$ (largest at $N = 137$); $|t^* - t_{\text{cf}}| \leq 0.00971 t^*$ for $q = \frac{1}{2}$, $10 \leq N \leq 200$ (largest at $N = 133$); (b) $d = 3$: $|t^* - t_{\text{cf}}| \leq 0.00275 t^*$ for $q = \frac{4}{5}$, $10 \leq N \leq 120$ and $N \in \{130, 140, 150, 160\}$; $|t^* - t_{\text{cf}}| \leq 0.00157 t^*$ for $q = \frac{1}{2}$, $10 \leq N \leq 80$. For $d = 2$, $(t_{\text{cf}} - t^*)/t^*$ is positive for $N \leq 13$ and negative for $N \geq 14$ on these ranges; for $d = 3$ and $q = \frac{4}{5}$ it is positive and falls, not monotonically, to 3.2×10^{-5} at $N = 160$. (Source: [R5, Thms 21, 43, Rems 22, 44], bases T1, T3 and T5.)*

Corollary 7.22 (Accuracy of the closed form for every $N \geq 10$; computer-assisted). *For $d = 3$, $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and every $N \geq 10$: $|t^* - t_{\text{cf}}| \leq 0.0030 t^*$. For $d = 2$: $|t^* - t_{\text{cf}}| \leq 0.0100 t^*$ for $q = \frac{4}{5}$ when $10 \leq N \leq 301$, $N \in \{350, 401, 450, 500, 600\}$ or $N \geq 1761450$, and for $q = \frac{1}{2}$ when $10 \leq N \leq 200$ or $N \geq 554846$. (Source: [R0, Cor 8.15], base T3 with T1, T5.)*

Conjecture 7.23. (a) For $d = 2$, $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and every $N \geq 10$, $|t^* - t_{\text{cf}}| \leq 0.0100 t^*$. (b) At $q = 1$, where the PMF is not unimodal in general, every global maximiser satisfies the laws of Theorem 7.20. (Source: [R5, Conj 48], [R3, Rem 7.19(ii)].)

Numerical observation 7.24 (The limit of the scaled error). $X \mu_1(t_c^* - qt_{\text{cf}})$ appears to converge, to 3.74 for $d = 2$ and to -0.22 for $d = 3$ (exact continuous-time modes up to $N = 1.7 \times 10^7$ and $N = 4096$, and the formal limit expression, which gives 3.7396 and -0.2217). The certified windows for $N \geq 10^{40}$ ($d = 2$), $(2.52, 4.65)$, and for $N \geq 10^6$ ($d = 3$), $(-0.83, 1.43)$, contain these values; at the largest computed sizes the quantity is 2.81 ($d = 2$) and -0.214 ($d = 3$). (Source: [R3, Rem 5.3(iii)].)

All continuous-time modes of Section 5 with $N \geq 12$ for $d = 2$ (40 sizes) and $N \geq 10$ for $d = 3$ (25 sizes) lie inside the windows of Theorem 7.18, and the 61 sizes with $N \geq 20$ inside the two-sided bounds of Theorem 7.19 (a check in double precision). The largest continuous-time error of the closed form in Table 4, 0.970% at $N = 113$ for $d = 2$, lies on the same plateau as the certified discrete-time maxima 0.970% ($N = 137$) and 0.971% ($N = 133$).

What remains open. In two dimensions the window of Theorem 7.18 is a few units wide, whereas the scaled error appears to converge; proving the limit needs $o(1)$ -accurate bounds on the pole data, and the 1% accuracy for $302 \leq N < 1\,761\,450$ at $q = \frac{4}{5}$ (apart from five certified sizes) and $201 \leq N < 554\,846$ at $q = \frac{1}{2}$ is Conjecture 7.23(a). In discrete time close to $q = 1$ nothing is proved in $d = 2, 3$ for $N < N_1(q)$, and $q = 1$ is Conjecture 7.23(b); in one dimension the crossover between the laws for $q < 1$ and $q = 1$ is open [R0, Section 12]. Unimodality for $\frac{4}{5} < q < 1$ and every N and log-concavity for $d \geq 2$ are Conjecture 7.12. For the centre placements only unimodality and bounds on the location of the continuous-time mode [R4, Cor 4.13] are proved, and nothing of this section is proved for lattices with blocked sites, apart from the certificate of Proposition 3.1.

8 Inert defects

Throughout this section $d = 2$, the start is the corner $\mathbf{o} = (0, 0)$, the target the opposite corner $\mathbf{a} = (N - 1, N - 1)$, $q = 0.8$, and $N = 35$ unless stated; the clean values are $T = 14\,100.8$ steps, $t^* = 2493$ steps and $t^*/T = 0.1768$. Times are in steps. Extended tables, the validation of the codes and the proofs are in Supplementary Section S9.

8.1 Blocked sites and exact computation

Let $\mathcal{B} \subset \Omega$ be the set of blocked sites, with $\mathbf{o}, \mathbf{a} \notin \mathcal{B}$, $M = |\mathcal{B}|$ and $p = M/N^2$. An attempted move that would leave Ω or land on a blocked site is cancelled. The walker can visit only the open cluster $\mathcal{C}$, the connected component of $\Omega \setminus \mathcal{B}$ that contains $\mathbf{o}$; we take $\mathcal{C}$ as the state space, write $n = |\mathcal{C}|$ and require $\mathbf{a} \in \mathcal{C}$. Open sites enclosed by blocked ones (pockets) are simply not in $\mathcal{C}$.

Proposition 8.1 (Mean first-passage time on the open cluster). *The walk in which every move onto a blocked site or off the lattice is cancelled has a symmetric, irreducible transition matrix on $\mathcal{C}$, and*

$$T_{\mathbf{o} \rightarrow \mathbf{a}} = \frac{2dn}{q} (G_{\mathbf{aa}} - G_{\mathbf{oa}}), \quad (41)$$

where G is the Moore–Penrose pseudo-inverse of the graph Laplacian of $\mathcal{C}$ with unit conductances.

Proof. Two neighbouring sites of $\mathcal{C}$ exchange probability $q/(2d)$ per step in both directions, and every cancelled attempt is added to the diagonal, so $I - P = \frac{q}{2d} \mathcal{L}_{\mathcal{C}}$ with $\mathcal{L}_{\mathcal{C}}$ the Laplacian of $\mathcal{C}$; Proposition 2.4 applies on $\mathcal{C}$. $\square$

Blocking sites lowers the number n of sites to be searched and, as a rule, raises the resistance factor $G_{\mathbf{aa}} - G_{\mathbf{oa}}$; the product need not increase. Every configuration is solved exactly: the mean by a sparse solve, the PMF by eigendecomposition of the symmetric matrix Q or by time stepping, and the mode by exhaustive search together with the certificate of Proposition 3.1, which uses only the symmetry of Q and applies unchanged on $\mathcal{C}$; here it is evaluated in double precision, without the a-priori round-off bound of Section 3 (Supplementary Section S9.1). Two implementations, written separately and sharing no code, agree with each other and with a direct simulation of the walker rule (Table S7); they produced two independent samples of placements, *sample A* and *sample B*.

8.2 Uniform and corner-thinned placement

We solved 400 placements at each of 13 densities, $p = 0.02$ to 0.35 , under two placement laws, in each of the two samples (20 800 configurations in all): *uniform* (every site except $\mathbf{o}$ and $\mathbf{a}$ equally likely) and *corner-thinned*, in which $M = \text{round}(pN^2)$ sites are sampled without replacement with weight 1 except in the 4×4 block of sites at each of the two corners, where the weight is $\exp[-(4 - \rho)/2]$ with ρ the Euclidean distance to that corner ($e^{-3/2}$ at the two neighbours of the corner, rising to $e^{0.12} \approx 1.13$ at the far corner of the block, where $\rho = 3\sqrt{2}$; the corner itself is never blocked). It is a simple placement law correlated with the positions of start and target: it thins the defects next to them and leaves the bulk uniform. Placements are redrawn until $\mathbf{a} \in \mathcal{C}$. The two samples agree: over the 26 cells the largest standardised difference between them is 1.9 for the mean, 2.1 for the mode and 1.9 for the ratio. Figure 3 shows the pooled results; Table S8 gives them in full. Two estimators of the ratio must be kept apart, the mean of the per-placement ratios, $\langle t^*/T \rangle$, and the ratio of ensemble means, $\langle t^* \rangle / \langle T \rangle$; between placements T fluctuates more than t^* (relative standard deviations 26% and 10% at $p = 0.2$ under uniform placement), and the first estimator lies above the second in every cell.

Uniform placement. Up to $p = 0.14$ the ratio is unchanged to within a few per cent, with a sign that depends on the estimator: $\langle t^* \rangle / \langle T \rangle$ lies between 0.8% below and 0.3% above the clean value, whereas $\langle t^*/T \rangle$ exceeds it by at most 2.4% for $p \leq 0.12$ and by 3.2% at $p = 0.14$; between 27% and 37% of the individual placements with $p \leq 0.2$ have a ratio *below* the clean one. From $p = 0.16$ both estimators rise, to 0.1926 ± 0.0022 (+8.9%) and 0.1851 (+4.7%) at $p = 0.2$. At $p = 0.2$ the mean and the mode are multiplied by 1.825 ± 0.033 and 1.911 ± 0.013 , while the coefficient of variation hardly moves ($0.918 \rightarrow 0.910$): the distribution is delayed, not reshaped.

Corner-thinned placement. Here $\langle t^*/T \rangle$ increases monotonically with p : 0.1794 ± 0.0003 , 0.1904 ± 0.0010 , 0.2066 ± 0.0020 and 0.2550 ± 0.0050 at $p = 0.02$, 0.10 , 0.20 and 0.35 . A least-squares line through the per-placement ratios for $p \leq 0.2$ has slope $+0.159 \pm 0.007$ per unit p , against $+0.081 \pm 0.009$ under uniform placement. At $p = 0.2$ the mean and the mode are multiplied by 1.606 ± 0.023 and 1.829 ± 0.011 . Section 8.4 explains the difference: thinning the defects next to the target, where the mean is most sensitive, slows the mean less than the mode.

8.3 Effective medium and time dilation

The walk with cancelled moves on a site-diluted lattice is the “blind ant” of the percolation literature (Havlin and Ben-Avraham, 1987). Its stationary distribution is uniform on the accessible cluster, and the Einstein relation ties the long-distance diffusivity D_{eff} to the conductivity Σ of the network of unit resistors on the open bonds: $\Sigma/\Sigma_0 = P_\infty D_{\text{eff}}/D_0$, where P_∞ is the fraction of sites in the accessible cluster (Havlin and Ben-Avraham, 1987; Stauffer and Aharony, 1994). We define the time-dilation factor

$$\Theta(p) = \frac{D_0}{D_{\text{eff}}(p)} = P_\infty(p) \frac{\Sigma_0}{\Sigma(p)}. \quad (42)$$

The hypothesis to be tested is that, on scales large compared with the percolation correlation length, uniformly random defects multiply every time scale by $\Theta(p)$ and leave the shape of the first-passage law, hence ϱ , unchanged. For the mean this is the coarse-grained form of (41). Θ diverges at the blocked fraction $p_c = 1 - 0.5927 = 0.4073$ (Newman and Ziff, 2000).

At small p the conductivity follows from the response of a single blocked site. Let $\mathcal{K}$ be the potential kernel of the planar simple random walk (Spitzer, 1976), with $\mathcal{K}(\mathbf{0}) = 0$, $\mathcal{K}(\pm \mathbf{e}_i) = 1$ and $\mathcal{K}(\pm 2\mathbf{e}_i) = 4 - 8/\pi$.

Proposition 8.2 (Dipole response of one blocked site). *On the infinite square lattice with unit conductances, remove the site $\mathbf{0}$ together with its four bonds and impose the potential $-Ex_1$ at infinity. Then: (a) the potential is unique, $U(\mathbf{x}) = -Ex_1 - \frac{\pi E}{8} [\mathcal{K}(\mathbf{x} + \mathbf{e}_1) - \mathcal{K}(\mathbf{x} - \mathbf{e}_1)]$, and at the two neighbours $\pm \mathbf{e}_1$ of the removed site it is $\mp \pi E/2$ (instead of $\mp E$); (b) the removed site*

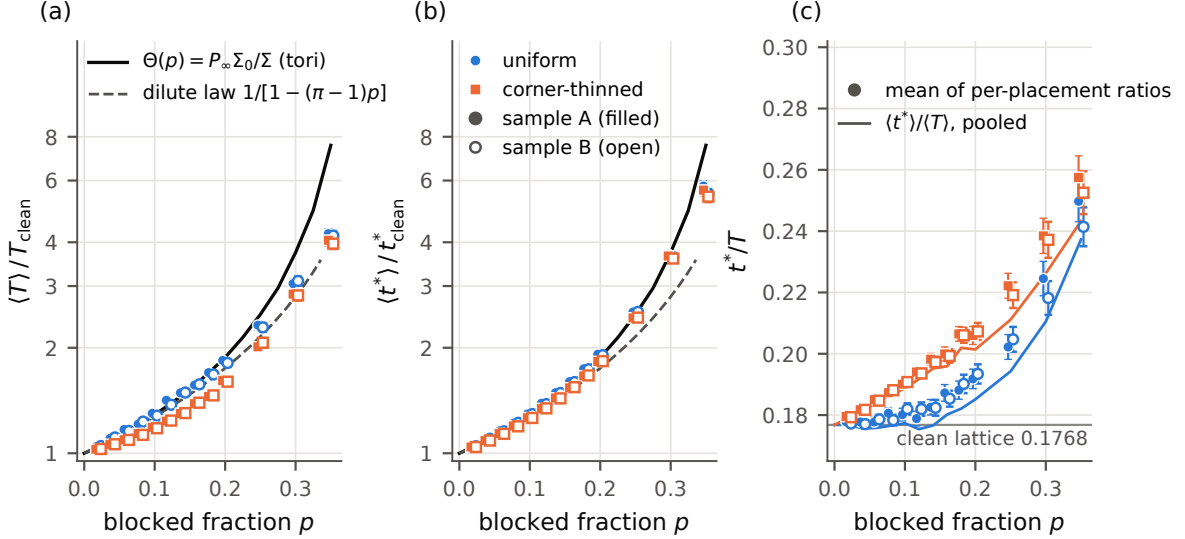


Figure 3: Density sweep, $N = 35$, $q = 0.8$, corner to opposite corner, discrete time. (a) Mean and (b) mode divided by their clean values, and (c) t^*/T , against the blocked fraction p , for uniform placement (circles) and corner-thinned placement (squares). Filled symbols: sample A; open symbols: sample B (400 placements each; bars are 95% intervals over placements; the samples are displaced horizontally for clarity). In (a) and (b) the solid curve is the homogenised factor $\Theta_{192}(p)$ measured on tori (Table S9) and the dashed curve the dilute law (43) in the form $1/[1 - (\pi - 1)p]$. In (c) the symbols are means of per-placement ratios and the lines the pooled ratio of ensemble means; the horizontal line is the clean value 0.1768.

lowers the dissipated power by πE^2 , in the sense that $\sum_b U_b^0 U_b = \pi E^2$ over the four removed bonds, and in every finite network of unit resistors with prescribed boundary potentials removing a set of bonds lowers the dissipated power by exactly $\sum_b \text{removed} U_b^0 U_b$.

Formal result 8.3 (Dilute law). For uniformly placed blocked sites on the square lattice, as $p \rightarrow 0$,

$$\frac{\Sigma}{\Sigma_0} = 1 - \pi p + O(p^2), \quad \frac{D_{\text{eff}}}{D_0} = 1 - (\pi - 1)p + O(p^2), \quad \Theta(p) = 1 + (\pi - 1)p + O(p^2). \quad (43)$$

The derivation treats each blocked site as isolated in the unperturbed field (Proposition 8.2 and $P_\infty = 1 - p + O(p^4)$); the step from one defect to a density of defects is not proved here, hence the label. The law is not new: Watson and Leath (1974) gave $\Sigma/\Sigma_0 = 1 - \pi p + \frac{1}{2}\pi p^2$ from an effective-medium theory, and Nieuwenhuizen et al. (1986) calculated the diffusion coefficient and conductivity of exactly this walk to order p^2 , with the coefficients printed in Ernst et al. (1987):

$$\frac{\Sigma}{\Sigma_0} = 1 - \pi p + 1.2858 p^2 + \dots, \quad \frac{D_{\text{eff}}}{D_0} = 1 - (\pi - 1)p - 0.8558 p^2 + \dots. \quad (44)$$

Our own evaluation of the pair term on tori gives 1.28588 and -0.85571 , within 10^{-4} of the printed values, and the conductivities measured on 192×192 tori agree with (44) to 6.2×10^{-4} for $p \leq 0.18$; the estimates Θ_{192} of the time-dilation factor are 1.045, 1.289, 1.877 and 3.739 at $p = 0.02, 0.1, 0.2$ and 0.3 (Supplementary Table S9).

Test at $N = 35$. Under uniform placement and $p \leq 0.2$ the pooled factor of the mean is within 2.8% of $\Theta_{192}(p)$ and that of the mode within 1.82%, with no fitted parameter (Fig. 3(a,b)); these are ensemble-mean estimates, whose 95% intervals reach $\pm 1.8\%$ for the mean and $\pm 0.72\%$ for

Table 7: Localised random placements of M blocked sites ($N = 35$, $q = 0.8$, 200 placements per row and sample, mean $\pm$ 95% confidence interval). Factors are relative to the clean lattice, for which $t^*/T = 0.1768$.

region	M	sample A			sample B
		T factor	t^* factor	t^*/T	t^*/T
anywhere	40	1.086 ± 0.017	1.080 ± 0.005	0.1771 ± 0.0016	0.1780 ± 0.0016
centre	40	1.0086 ± 0.0008	1.150 ± 0.003	0.2015 ± 0.0004	0.2014 ± 0.0004
near start	10	0.9933 ± 0.0001	1.012 ± 0.001	0.1801 ± 0.0001	0.1802 ± 0.0002
near start	40	0.999 ± 0.003	1.320 ± 0.027	0.2329 ± 0.0040	0.2368 ± 0.0048
near target	10	1.239 ± 0.029	1.056 ± 0.006	0.1530 ± 0.0020	0.1530 ± 0.0019
near target	20	1.659 ± 0.057	1.134 ± 0.009	0.1249 ± 0.0024	0.1276 ± 0.0025
near target	30	2.281 ± 0.083	1.230 ± 0.011	0.0994 ± 0.0023	0.0991 ± 0.0024
near target	40	3.367 ± 0.127	1.357 ± 0.015	0.0752 ± 0.0022	0.0741 ± 0.0021

the mode (relative to Θ_{192}), so that the deviations of the mean are of the size of their sampling uncertainty. For $p \geq 0.25$ the mean falls below Θ (3.08 against 3.74 at $p = 0.3$) while the mode still follows it (3.66); those densities describe a selected ensemble on a lattice not much larger than the correlation length. The simplest mean-field description of defects, a reduced activity $q_{\text{eff}}(p)$, cannot capture the changes of ϱ found below: a change of activity only rescales time, and by (4) and Proposition 2.2(d) it leaves ϱ unchanged up to a few steps (Section 6).

Larger lattices. For uniform placement at $p = 0.1$ and 0.2 we computed modes up to $N = 200$, certified by Proposition 3.1 in double precision, and means up to $N = 280$, with the most precise estimates $\Theta_{200}(0.1) = 1.2877$ and $\Theta_{200}(0.2) = 1.8783$. The mode factor equals Θ to within 1.11% at $p = 0.1$ and 1.6% at $p = 0.2$ for every N from 15 to 200; the factor of the mean divided by Θ , averaged over $N = 35$ –280, is 1.004 ± 0.005 and 0.987 ± 0.007 . The shift of $\langle t^*/T \rangle$ relative to the clean lattice of the same size is between +0.6% and +1.2% for $N \geq 50$ at $p = 0.1$; at $p = 0.2$ it falls from $+9.4 \pm 1.8\%$ at $N = 35$ (in the 400 placements of this size study; the 800 pooled placements of Section 8.2 give +8.9%) to $+4.5 \pm 1.4\%$ at $N = 100$ and $+3.9 \pm 2.7\%$ at $N = 200$: it halves but is not shown to vanish (Supplementary Table S10). A slow approach is to be expected, because in (41) the constant that accompanies $\ln N$ in G_{aa} is set by the immediate surroundings of the target.

Conjecture 8.4 (Homogenisation). *Fix $p < p_c$ and place $\text{round}(pN^2)$ blocked sites uniformly, conditioned on $\mathbf{a} \in \mathcal{C}$. As $N \rightarrow \infty$, $\langle T \rangle / (\Theta(p) T_{\text{clean}}) \rightarrow 1$ and $\langle t^* \rangle / (\Theta(p) t_{\text{clean}}^*) \rightarrow 1$, so that $\langle t^* \rangle / \langle T \rangle$ divided by the clean ratio at the same N tends to one.*

8.4 Placement decides the direction

By (41), $T = (4n/q)(G_{\text{aa}} - G_{\text{oa}})$. The rows of G sum to zero, so $(4n/q)G_{\text{aa}}$ is the mean first-passage time from a uniformly distributed start, and the actual start enters only through the cross term G_{oa} (−0.221 against $G_{\text{aa}} = 2.082$ on the clean lattice). We call G_{aa} the *capture resistance* of the target. It controls the mean: over the 400 uniform placements at $p = 0.1$ and at $p = 0.2$ the rank correlation between T and G_{aa} is 0.9995 and 0.9986 (sample A; 0.9995 and 0.9982 in sample B), whereas that between T and the number of blocked sites near the start is below 0.08 in absolute value; and at $p = 0.2$ the uniform and the corner-thinned placements leave the same accessible volume ($n/N^2 = 0.797$) but raise $G_{\text{aa}} - G_{\text{oa}}$ by the factors 2.310 and 2.022. The mode responds to a different part of the lattice. Blocking a single site shortens the mean for 67.6% of the sites (the lost volume), and the positive responses are concentrated around the target, where 64% of $\sum_{\mathbf{x}} \chi_T(\mathbf{x})$ comes from sites within distance 3; the response of the mode is spread over the whole region between start and target (19% within distance 3 of the target), and only 5.4% of the sites lower it (Fig. S14).

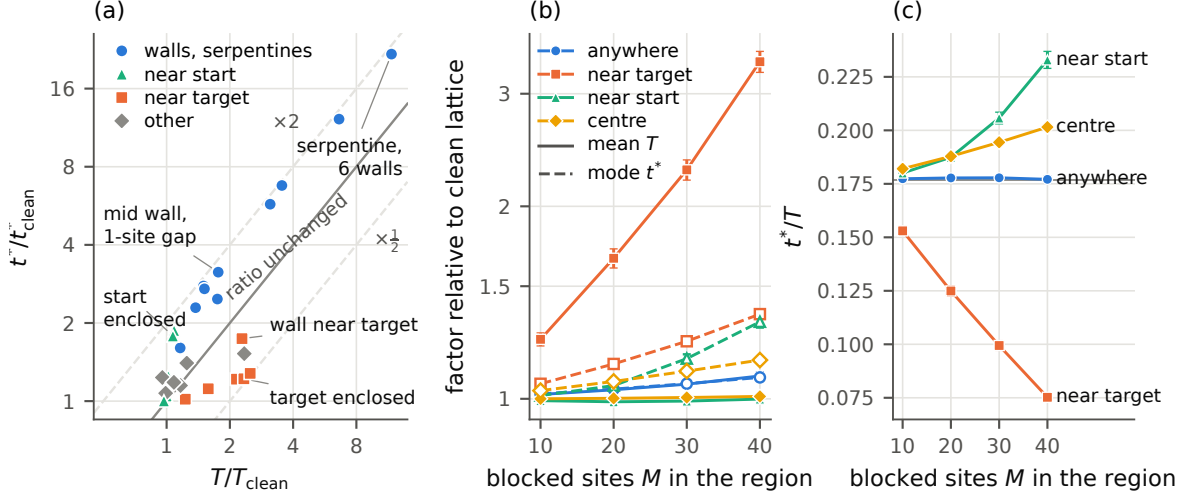


Figure 4: Placement decides the direction ($N = 35$, $q = 0.8$, corner to opposite corner, discrete time). (a) Mode factor against mean factor for the 28 deterministic geometries of Table S19, logarithmic axes; on the solid line t^*/T keeps its clean value, on the dashed lines it is doubled or halved. (b) Mean (solid lines) and mode (dashed lines) relative to the clean lattice for M blocked sites drawn inside a region. (c) The resulting ratio; the grey line is the clean value. In (b) and (c) the points are means over the 200 placements of sample A and the bars 95% confidence intervals.

To separate placement from density we block M sites drawn uniformly inside one region: within Euclidean distance 12 of the target (122 candidate sites), within distance 12 of the start (122 sites), within 6.2 of the centre (121 sites), or anywhere (Table 7, Fig. 4). Forty blocked sites, 3.3% of the lattice, multiply the mean by 3.37 ± 0.13 and the mode by 1.36 near the target, so that the ratio falls to 0.0752 ± 0.0022 , less than half its clean value; near the start they leave the mean unchanged ($\times 0.999 \pm 0.003$), delay the mode by a third and raise the ratio to 0.2329 ± 0.0040 . Deterministic obstacles behave in the same way: serpentine corridors push the law towards the one-dimensional shape ($t^*/T = 0.316\text{--}0.339$), walls between start and target raise the ratio to between 0.244 and 0.328, whereas a wall next to the target lowers it to 0.135; enclosing the target makes the law more nearly exponential (ratio 0.0925 for ten blocked sites), and a solid 12×12 block in the centre *lowers* the mean by 4.6% and delays the mode by 23%. Hence t^*/T is not a function of the defect density: it decreases when the defects surround the target, increases when they lie around the start or between start and target, and, when they are spread uniformly, stays within a few per cent of its clean value at low density ($p \leq 0.14$; Section 8.2) but rises by 4.7% to 8.9%, depending on the estimator, at $p = 0.2$.

Two time scales. Beyond the transit time the PMF is dominated by its two slowest terms, with time constants $T_{\text{cap}} = -1/\ln(1 - q\nu_0)$ (the capture time) and $\tau_1 = -1/\ln(1 - q\nu_1)$; with $1/\tau = 1/\tau_1 - 1/T_{\text{cap}}$ and $c = -b_1/b_0$, the maximiser of the two-term sum is $t^* \approx \tau \ln[c(1 + T_{\text{cap}}/\tau)]$. A pure dilation multiplies T_{cap} and τ by the same factor and leaves t^*/T unchanged, whereas a perturbation that raises only the capture time moves the mode sub-linearly: on the clean lattice the exponent $\partial \ln t^*/\partial \ln T_{\text{cap}}$ is 0.21 with τ and c held fixed, and 0.17 with τ_1 and c held fixed (τ then follows T_{cap}). The exact elasticities $\ln(t^* \text{ factor})/\ln(T \text{ factor})$ of obstacles that enclose the target are somewhat larger, 0.234 to 0.310, and the regression of $\ln t^*$ on $\ln T$ across the uniform placements at $p = 0.2$ has slope 0.30. This heuristic accounts for the sign and the approximate size of the effect and has no controlled error; it fails for a wall near the target, which also lengthens the transit (Supplementary Section S9.8).

A one-parameter family of shapes. Over all 13631 exactly solved configurations of sample A the coefficient of variation is nearly a function of ρ alone (Pearson correlation -0.981 ; a quadratic

fits with a root-mean-square residual of 0.0045); sample B gives -0.983 . We record this as an empirical regularity, without a theory: size, density and placement appear to act mainly by moving the law along one line between a transit-limited shape and a capture-limited, nearly exponential one.

Other defect types. On a chain, a barrier crossed with reduced probability costs more the nearer it is to the target (Proposition S9.1; for a single barrier this is Eq. (12) of Sarvaharman and Giuggioli (2023), who also showed that a barrier behind the start changes the mode and the tail but not the mean, their ‘disorder indifference’), and uniformly random barriers are again a dilation of time. On the square lattice, permeable obstacles at $p = 0.1$ leave the ratio within 3.2% of its clean value, but the limit of vanishing permeability is singular, because a site that can be entered at all carries its full share of the stationary distribution. Partial traps, which remove the walker with probability Δ per step, remove the long trajectories and act in the opposite sense to obstacles: the ratio of mode to conditional mean rises from 0.178 to 0.638 as Δ grows from 10^{-5} to 10^{-2} at $p = 0.1$ (Supplementary Section S9.10).

Within this model, and at the sizes computed, the mean and the mode respond to different parts of the lattice: the mean to the capture resistance of the target, which is set by its immediate surroundings, and the mode to the transit between start and target. We make no claim beyond the lattices, sizes and defect models examined here.

9 Discussion and conclusions

9.1 What is established, and with what status

The results of this article do not all have the same standing; Table 8 sorts them. “Exact numerics” are computations that are exact in the sense of Section 3 (up to double-precision round-off for time stepping, to a measured accuracy for Laplace inversion) and cross-checked by separately written routes and implementations; they hold at the sizes run and are not theorems. The proofs of Section 7 were checked within this project (Section 7.1), not by external referees.

Table 8: Status of the main statements. Discrete-time values are for $q = 0.8$; defect results are for $d = 2$ with a corner start and the opposite corner as target.

Status	Statements	Where
Proved	Exact structure, interlacing, pseudo-inverse formula; cosine sum in any dimension; single sums for $d = 2$ and every $N \geq 2$; $qT_N = 2N^2R_N$; double sum for $d = 3$; the large- N expansion; certificate of the global mode; the limit $t^*/T \rightarrow 2\xi^*$ on the chain in continuous time and for $q \leq \frac{1}{2}$; factorisation of the renewal ratio; unimodality of the three point-target placements in continuous time and for $q \leq 1/(1 + \cos \frac{\pi}{N})$; signs of the residues (their non-alternation for $d \geq 2$, $N \geq 3$); bounds on the slowest pole; failure of unimodality near $q = 1$ on the chain; resistance identity and dipole response with blocked sites.	Secs. 2–8

Status	Statements	Where
Proved with computer assistance	Unimodality for $q \leq \frac{4}{5}$ from corner to corner (for the centre placements when $d \geq 2$; a threshold attained on the chain with $N = 4$), and on finite ranges beyond: up to $q = 1$ for $d = 3$ ($N \leq 30$), $d = 4$ ($N \leq 8$) and $d = 5$ ($N \leq 5$), and $q \leq \frac{499}{500}$ for $d = 2$ ($N \leq 64$). In one dimension $t^*/T \rightarrow 0.33328426549\dots$ for every q , with windows of width about one step for $q \leq \frac{4}{5}$. For corner-to-corner transport in $d = 2, 3$: the scaled error of the closed form (35) bounded for every $N \geq 12$ ($d = 2$) and $N \geq 10$ ($d = 3$), and the leading laws with explicit remainders, in continuous time and, in discrete time, for $q \leq 0.99$ once $N \geq N_0$ and for every $q < 1$ once $N \geq \max(N_0, N_1(q))$; exact modes on finite ranges, the closed form within 0.971% ($d = 2$) and 0.275% ($d = 3$) there for $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and $N \geq 10$, and within 0.3% in three dimensions for $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and every $N \geq 10$. Means: a remainder for every N ($d = 2$); K_3, K'_3 and the next constant ($d = 3$).	Secs. 4, 7
Exact numerics at the stated sizes	Distributions and modes in discrete time up to $N = 1000, 401, 100$ ($d = 1, 2, 3$), each mode a certified global maximiser (in exact arithmetic or with an a-priori bound on round-off), and in continuous time up to $N = 1.7 \times 10^7$ ($d = 2$) and $N = 4096$ ($d = 3$); for $d = 2$, t^*/T falling from 0.27 ($N = 5$) to 0.056 ($N = 1.7 \times 10^7$), for $d = 3$ from 0.153 ($N = 5$) to 0.0118 ($N = 100$) and 4.1×10^{-4} ($N = 4096$, continuous time); the expression of K'_3 through ξ_H and $\varkappa$; all defect statistics, up to placement sampling error: under uniform placement t^*/T within 3.3% of its clean value 0.1768 up to $p = 0.14$, and at $p = 0.2$ the mean of the per-placement ratios 8.9% above it (the ratio of ensemble means 4.7%); forty blocked sites near the target give 0.075, near the start 0.233; the coefficient of variation is nearly a function of t^*/T (an empirical regularity).	Secs. 3–8
Formal asymptotics	Closed-form approximation (34) for placements other than corner to corner (corner to centre: within 0.964% for $d = 2$ and 0.206% for $d = 3$ at the computed sizes); dilute law (43); first-order susceptibility estimate.	Secs. 6, 8
Heuristic	Two-time-scale account of the sub-linear response of the mode to a change of capture time.	Sec. 8
Conjecture	Time dilation by $\Theta(p)$ exact as $N \rightarrow \infty$; the closed form within 1% in two dimensions for every $N \geq 10$; the mode laws at $q = 1$; unimodality for $\frac{4}{5} < q < 1$ and every N ; log-concavity for $d \geq 2$.	Secs. 7, 8
Refuted	Alternating residues for opposite corners in $d \geq 2$; unimodality of every point-target law; the closed form in one dimension (on the certified ranges with $N \geq 10$ at $q = \frac{4}{5}$ it overestimates the mode by at least 15.77% and at most 22.51%).	Sec. 7

9.2 The physical picture

Two times govern the problem. The mean is a capture time: a walker that has spread over the box finds the target at a rate fixed by the number of sites times the capture resistance of the target, (13), which gives N^2 , $N^2 \ln N$ and N^3 for $d = 1, 2, 3$. The mode is a relaxation time: the first-passage density peaks when the transient of arrival across the box, which decays at a rate $\mu \propto N^{-2}$ in every dimension, has fallen to the level of the slow capture decay, and this costs only the logarithmic factor $\ln(AX)$ of (34), where $X = \mu qT$ measures how weak a sink the target is. For corner-to-corner transport in two and three dimensions this picture is a theorem (Section 7.5); in one dimension X stays bounded and the two times are of the same order. Hence the dimension dependence of Table 3: t^*/T tends to a constant for $d = 1$, falls slowly for $d = 2$

and quickly for $d = 3$, and the separation between mean and mode grows with dimension. For a point target in $d \geq 2$ the law is close to an exponential preceded by a short dead time, and the mode marks the end of that dead time: it is not a typical time. The appropriate typical value is the median, about $0.7T$, and statements about waiting times must be made with the survival function $S(t)$, not with t^*/T .

The same two times organise the defect results. Uniformly placed blocked sites slow capture and relaxation by nearly the same factor $\Theta(p)$, so that t^*/T stays near its clean value at low density; the rise at $p = 0.2$ shrinks with N but is not shown to vanish. Blocked sites around the target raise the capture resistance, hence the mean, far more than the mode; blocked sites around the start or between start and target delay the transit, hence the mode, and leave the mean almost unchanged.

9.3 Relation to the literature

By the commute-time identity the two-dimensional results on the mean are statements about the resistor grid, and in that form they were already published (Condamin and Bénichou, 2006; Wu, 2004; Giuggioli, 2020; Essam and Wu, 2009; Izmailian and Huang, 2010; Tan and Tan, 2020); we add a self-contained proof in the random-walk setting for both parities of N and record the overflow-free form (18), an elementary rearrangement. The shape of the law agrees with the picture of geometry-controlled kinetics in confined domains (Bénichou et al., 2010; Meyer et al., 2011; Bénichou and Voituriez, 2014; Mattos et al., 2012; Godec and Metzler, 2016; Grebenkov et al., 2018) and with the exponential approximation of hitting times for reversible chains (Aldous and Brown, 1992); we add exact lattice distributions and modes and their explicit dependence on N and d ; the most probable time of direct trajectories and the relation of the median to the mean are given by Godec and Metzler (2016) and Grebenkov et al. (2018). Apart from their classical ingredients (the interlacing of relaxation and first-passage spectra (Hartich and Godec, 2018, 2019), the mixture and exponential laws from the uniform and the quasi-stationary start (Aldous and Fill, 2002), unimodality on the chain in continuous time (Keilson, 1971)), we have not found in print the theorems of Section 7 on the mode of a lattice first-passage time, the explicit closed-form approximation (34), whose ingredients are the lattice counterpart of known small-target asymptotics (Ward and Keller, 1993; Isaacson and Newby, 2013), or the expressions (25) through the lemniscate constant, which specialise the general expressions of Izmailian and Huang (2010). Our search was targeted, not exhaustive, and we claim no priority. The dilute law for blocked sites is in the literature (Watson and Leath, 1974; Nieuwenhuizen et al., 1986; Ernst et al., 1987), and our measured conductivities agree with its published second-order form. That the placement of a defect decides whether the mean or only the mode and the tail respond was shown by Sarvahaarman and Giuggioli (2023) for a single barrier on a chain (their Eq. (12) is our Proposition S9.1 for one barrier); Section 8 adds the exact two-dimensional ensembles of blocked sites.

9.4 Limitations

1. Unimodality is proved for the three point-target placements studied here in continuous time, and in discrete time for $q \leq \frac{4}{5}$ from corner to corner (for the centre placements when $d \geq 2$), and on finite ranges beyond; two computed runs at $q = 0.9$ lie outside the proved ranges of unimodality, and the 16 computed corner-to-corner runs at $q = 0.3$ and 0.9 lie outside the certified ranges of the mode and rest on the a-priori round-off bound of Section 3. The continuous-time modes at sizes beyond time stepping are global maximisers by Theorem 7.6, but their location rests on numerical Laplace inversion, to the accuracy of Table S2, not on a certificate.
2. The closed form is proved only for corner-to-corner transport in $d = 2, 3$, with a window a few units wide in $X\mu_1(t_c^* - qt_{cf})$. In two dimensions its 1% accuracy is proved only on

finite ranges and for very large N , and the asymptote $N^2 \ln \ln N$, although proved, is not reached by the data.

3. The computer-assisted proofs rest on their trusted bases (Python integers; Arb ball arithmetic through python-flint together with the transcription of the formulas into the scripts; beyond the Python-integer ranges, a C program run twice, which is the only witness for 2242 of its 3824 certificates). No program was verified in a proof assistant. The certificate of Theorem 7.20(c) has been regenerated with the released ball-arithmetic implementation; it has not been independently reimplemented.
4. The expression of K'_3 through ξ_H and $\varkappa$ is identified numerically and is unproved.
5. Only four placements were studied (corner to corner, corner to centre, centre to corner, and exit through an absorbing boundary layer); (34) does not apply to the third.
6. Inert defects were studied only for $d = 2$, a corner target and $q = 0.8$, up to $N = 280$ for means and $N = 200$ for modes; barriers on the chain only at $N = 100$. A corner target has two neighbours, which makes the mean unusually sensitive to its surroundings. Placements are conditioned on start and target being connected; for $p \geq 0.3$ at $N = 35$ the results describe a selected, near-critical ensemble, not bulk properties.
7. Time dilation by $\Theta(p)$ is accurate at $p = 0.1$ and a 1–3% approximation at $p = 0.2$ at the sizes computed; its exactness as $N \rightarrow \infty$ is a conjecture. The model is a lattice walk with a single walker.

9.5 Open problems

(a) Unimodality for $\frac{4}{5} < q < 1$ and every N , and log-concavity of the law for $d \geq 2$. (b) Sharp constants for the closed form: N -dependent window constants sharp enough to close the gap $302 \leq N < 1\,761\,450$ of Conjecture 7.23(a), the value of the limit of $X\mu_1(t_c^* - qt_{cf})$, and the discrete walk close to $q = 1$. (c) The mode for a general start and target, including placements for which the slowest target-visible mode does not have a positive amplitude A . (d) A proof of the expression of K'_3 through ξ_H and $\varkappa$, and whether the series (24) diverges. (e) Defects in three dimensions and around interior targets, and the large- N behaviour under uniform placement (Conjecture 8.4).

9.6 Conclusions

For a lazy walk in a reflecting box with one absorbing site the mean first-passage time is given by exact finite sums and is equivalent, in two dimensions, to a resistance that was already in the literature. The most probable first-passage time obeys a different law: it follows the relaxation time of the box, not the capture time, and it separates from the mean more strongly as the dimension increases. For corner-to-corner transport this is proved, partly with computer assistance: the law is unimodal in continuous time and, in discrete time, for $q \leq \frac{4}{5}$, while unimodality fails near $q = 1$ on the chain and, at $q = 1$, at some sizes in two dimensions; in two and three dimensions its mode follows a parameter-free closed form with a bounded scaled error, and the leading laws $N^2 \ln \ln N$ and $N^2 \ln N$ hold with explicit remainders, in continuous time and, in discrete time, for $q \leq 0.99$ and for every $q < 1$ once N is large enough; in one dimension t^*/T tends to 0.33328 for every activity. In the square-lattice setting studied here (corner target, mainly $N = 35$), the direction in which inert defects move the ratio of the two times is set by where they sit, not by how many there are.

Data and code availability

All code and data needed to reproduce every number, table and figure of this article and of its Supplementary Material are in the repository (Zhouyi, 2026g)

<https://github.com/zhouyi-xiaoxiao/master-dissertation>,

together with the L^AT_EX sources and a file `reproduce.md` that lists the exact commands. The folder `proofs/` contains the companion notes R0–R5 (Zhouyi, 2026a,b,c,d,e,f) with the complete proofs of the results of Sections 4 and 7; the certificates of the computer-assisted proofs and the programs that check them are in `data/msc_rigorous_*` and `code/msc_rigorous_*`. All computations are deterministic or use fixed random seeds. The submission verification commands and their coverage are recorded in `submission/SCIENTIFIC-CHECKS.md`.

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